Computer Systems (SCS 1305 / IS 1202)

Short Note

Contents

1 Data Representation

1.1 Quantitative and Qualitative Data

Data representation is concerned with how numbers, characters and symbols are represented inside a computer.

- **Quantitative data** can be counted or measured and expressed using numbers. It may be integer or non-integer, and signed or unsigned.
- **Qualitative data** is descriptive/conceptual, and is categorised based on traits or characteristics rather than numeric value.

1.2 Bits, Bytes and Words

- A **bit** (binary digit) is the smallest unit of data in a computer; it holds either `0` or `1`.
- A **byte** is 8 consecutive bits, capable of storing a single character.
- A **word** is the natural, fixed-size unit of data handled as a whole by a particular processor's instruction set / hardware. The number of bits in a word (word size / word length) is a defining characteristic of a processor architecture.

Storage hierarchy (all base 1024, i.e. 2^{10}):

8 bits	= 1 Byte
1024 Bytes	= 1 Kilobyte (KB)
1024 KB	= 1 Megabyte (MB)
1024 MB	= 1 Gigabyte (GB)

1.3 Number Systems

System	Alphabet (allowed digits)
Decimal (base 10)	$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
Octal (base 8)	$\{0, 1, 2, 3, 4, 5, 6, 7\}$
Hexadecimal (base 16)	$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F\}$
Binary (base 2)	$\{0, 1\}$

A **leading zero** is any `0` digit that appears before the first non-zero digit in a positional number; leading zeros are not significant – they are simply placeholders (e.g. the "007" in James Bond conveys nothing extra numerically).

1.3.1 Decimal ↔ Binary

Decimal to binary: repeatedly divide the decimal number by 2, recording the remainder at each step, until the quotient is 0. The binary result is the sequence of remainders read from *last to first* (the last non-zero quotient's remainder is the most significant bit).

Example: Convert $(123)_{10}$ to binary.

$$123 \div 2 = 61 \text{ r } 1, \quad 61 \div 2 = 30 \text{ r } 1, \quad 30 \div 2 = 15 \text{ r } 0, \quad 15 \div 2 = 7 \text{ r } 1,$$

$$7 \div 2 = 3 \text{ r } 1, \quad 3 \div 2 = 1 \text{ r } 1, \quad 1 \div 2 = 0 \text{ r } 1$$

Reading the remainders from bottom (last) to top (first) gives $(123)_{10} = (1111011)_2$.

Binary to decimal: multiply each bit by 2 raised to the power of its position (position 0 is the least significant / rightmost bit), and sum the results.

Example: Convert 10011010_2 to decimal.

Bit position	7	6	5	4	3	2	1	0
Bit value	1	0	0	1	1	0	1	0
Place value	128	64	32	16	8	4	2	1

$$128 + 16 + 8 + 2 = 154 \quad \Rightarrow \quad 10011010_2 = 154_{10}$$

A useful alternative technique for binary to decimal conversion: work from the most significant bit, and at each position ask whether that power of two "fits" into what remains of the number, placing a 1 or 0 accordingly and subtracting if it fits, then continuing to the next lower power of two (this is effectively the same process worked in reverse, useful for converting decimal into binary directly by comparison against powers of two).

1.3.2 Hexadecimal Notation and Binary ↔ Hex

Each hex digit corresponds to exactly 4 bits (a "nibble"):

HEX	Bit Pattern	HEX	Bit Pattern
0	0000	8	1000
1	0001	9	1001
2	0010	A	1010
3	0011	B	1011
4	0100	C	1100
5	0101	D	1101
6	0110	E	1110
7	0111	F	1111

Binary to hexadecimal: group the binary digits into sets of 4 starting from the least significant bit (pad with leading zeros on the left if necessary), then convert each group of 4 bits to its hex digit.

Examples:

$$10010110_2 \;\Rightarrow\; 1001\;\; 0110 \;\Rightarrow\; 9\;\; 6 \;\Rightarrow\; 96_{16}$$

$$11011011_2 \;\Rightarrow\; 1101\;\; 1011 \;\Rightarrow\; D\;\; B \;\Rightarrow\; DB_{16}$$

$$0010100111110101_2 \;\Rightarrow\; 0010\;1001\;1111\;0101 \;\Rightarrow\; 2\;9\;F\;5 \;\Rightarrow\; 29F5_{16}$$

Hexadecimal to binary: reverse the process – expand each hex digit into its 4-bit binary pattern and concatenate.

Decimal ↔ hexadecimal / octal: convert via binary (decimal → binary → hex/octal), or repeatedly divide by 16 (hex) / 8 (octal) and read the remainders from last to first, analogous to the decimal-to-binary method.

Binary ↔ octal: group bits into sets of 3 (each octal digit corresponds to 3 bits), starting from the least significant bit, and convert each group.

Note: the reason grouping works is that $16 = 2^4$ and $8 = 2^3$, so each hex/octal digit exactly represents a fixed-width group of binary digits.

1.4 Character Representation

1.4.1 ASCII

ASCII (American Standard Code for Information Interchange) uses 7-bit patterns to represent:

- Letters of the English alphabet: `a--z` and `A--Z`
- Digits `0--9`
- Punctuation symbols: `( ) [ ] { } ' " ! / \`
- Arithmetic operator symbols: `+ - * < > =`
- Special symbols: space, `% $ # & @ ^`

With 7 bits, $2^7 = 128$ characters can be represented; the 8th bit was historically used as a **parity bit** for simple error checking. As reliability of hardware improved, this need for a parity bit faded, and manufacturers extended ASCII using the freed 8th bit to represent an additional 128 special characters (codes 128 to 255, i.e. 2^7 to $2^8 - 1$) – this is known as **Extended ASCII**.

Uppercase and lowercase versions of the same letter differ by exactly `0x20` (e.g. `A` = `0x41`, `a` = `0x61`).

1.4.2 Unicode

ASCII and EBCDIC are built around the Latin alphabet and cannot represent non-Latin scripts well, so many countries historically developed their own incompatible codes for native languages. **Unicode** solves this by using a much larger code space:

- Originally a 16-bit system, giving $2^{16} = 65{,}536$ possible characters, and is **downward compatible** with ASCII and Latin-1.
- As of the standard's growth (e.g. Unicode 13.0, March 2020), it defines over 140,000 characters covering 150+ modern and historic scripts plus symbol sets and emoji.
- Unicode version 3.0 (September 1999) added support for **Sinhala**.
- Characters are grouped into **blocks**; e.g. the Sinhala block occupies code page `0D80` to `0DFF` (128 code points).
- The Unicode Consortium was incorporated in California in January 1991.
- Unicode supports three encoding forms that all use the same character repertoire: **UTF-8**, **UTF-16** and **UTF-32**.

Example: the ASCII code for 'A' is `0x41`; the Unicode code point for 'A' is written `U+0041` (i.e. `00 41` in the Basic Latin block), showing the downward compatibility with ASCII.

2 Computer Arithmetic

2.1 Binary Addition, Subtraction, Multiplication, Division

Binary addition rules:

$$0 + 0 = 0, \qquad 0 + 1 = 1, \qquad 1 + 0 = 1, \qquad 1 + 1 = 10 \text{ (write } 0, \text{ carry } 1)$$

Example:

$$\begin{array}{r} {\scriptstyle 11111} \\ 01101 \\ +\ 10111 \\ \hline 100100 \end{array}$$

Binary subtraction rules:

$$0 - 0 = 0, \qquad 0 - 1 = 1 \text{ (with borrow)}, \qquad 1 - 0 = 1, \qquad 1 - 1 = 0$$

Example:

$$\begin{array}{r} {\scriptstyle ***}\ \text{(borrows)} \\ 101101 \\ -\ 010111 \\ \hline 010110 \end{array}$$

Binary multiplication: follows the same shift-and-add process as decimal long multiplication – the multiplicand is multiplied by each bit of the multiplier in turn, each partial product is shifted one place left, and the partial products are summed.

$$\begin{array}{rr} 1011 & \text{(multiplicand)} \\ \times\ 1010 & \text{(multiplier)} \\ \hline & 0000 \\ & 1011 \\ & 0000 \\ & 1011 \\ \hline & 1101110 \end{array}$$

Binary division: follows the same long-division process as decimal division, except at each step the quotient digit can only be 0 or 1.

2.2 Representing Numbers: Problems and Approaches

Any scheme for representing numbers in binary must deal with:

- How to represent both positive and negative numbers.
- Where the radix (binary) point goes, for non-integers.
- The range of values that can be represented in a fixed number of bits.

Three broad approaches exist:

- **Unsigned representation** – for non-negative integers only.
- **Signed representation** – for integers (positive and negative).
- **Floating-point representation** – for numbers with fractional parts.

Format for an n-bit unsigned number: all n bits, $b_{n-1} \ldots b_1 b_0$, form the magnitude, with b_{n-1} as the most significant bit (MSB).

Format for an n-bit signed number: bit b_{n-1} is the sign bit (0 = positive, 1 = negative) and the remaining $n-1$ bits, $b_{n-2} \ldots b_1 b_0$, form the magnitude.

2.2.1 Unsigned Binary Numbers

- Only positive numbers (including zero) can be stored.
- Smallest 8-bit value: `00000000` $= 0$.
- Largest 8-bit value: `11111111` $= 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255 = 2^8 - 1$.
- General range for n bits: 0 to $2^n - 1$ (i.e. 2^n distinct values).

2.2.2 Signed-Magnitude Representation

- The leftmost (most significant) bit is a dedicated **sign bit**: 0 = positive, 1 = negative. The remaining bits store the magnitude, unchanged, as for unsigned numbers.
- Smallest positive 8-bit value: `00000000` = 0.
- Largest positive 8-bit value: `01111111` = $64 + 32 + 16 + 8 + 4 + 2 + 1 = 127 = 2^7 - 1$.
- General range for positive numbers, n bits: 0 to $2^{n-1} - 1$.
- **Problem 1 – two representations of zero**: `00000000` (+0) and `10000000` (−0) both represent zero, which is ambiguous.
- **Problem 2 – arithmetic is awkward**: both sign and magnitude must be considered explicitly when adding/subtracting; naive bit-wise addition of the sign bits along with magnitude bits gives wrong answers. *Example:* $5 + (-3)$ using 8-bit sign-magnitude: $00000101 + 10000011 = 10001000$, which if read again as sign-magnitude means −8 – clearly wrong (the correct answer is 2). This shows that simple bit-wise addition does not work correctly for sign-magnitude numbers.

2.2.3 One's Complement

- Positive numbers are unchanged from their unsigned form (and still start with a 0).
- A negative number is represented by **inverting every bit** (0s become 1s and 1s become 0s) of the corresponding positive number – e.g. one's complement of $(10010100)_2$ is $(01101011)_2$ is not quite the right pairing; more precisely, to get $-x$, take the positive representation of x and flip every bit.
- Since positive numbers always start with 0, their complements (the negative numbers) always start with 1.
- For an n-bit one's complement number, the representable range is $-(2^{n-1} - 1) \leq D \leq +(2^{n-1} - 1)$, e.g. a 4-bit one's complement number ranges from −7 to +7 (with two representations of zero).
- **Subtraction using one's complement** requires an **end-around carry**: if the addition produces a carry out of the most significant bit (a 9th bit for an 8-bit system), that carry bit is added back into the least significant bit of the result. *Example:* $5 + (-3)$ in 8-bit one's complement:

$$00000101 + 11111100 = 1\,00000001 \text{ (9-bit result, carry = 1)}$$

$$00000001 + 1 \text{ (carry)} = 00000010 = +2 \quad \checkmark$$

- **Benefits**: easy to negate (just flip all bits); subtraction can be performed as addition.
- **Problems**: still has two representations of zero (`00000000` and `11111111`); requires the extra end-around-carry step, which is not "clean".

2.2.4 Two's Complement

- Positive numbers are unchanged from their unsigned form.
- A negative number is obtained by taking the one's complement (bit-flip) of the corresponding positive number and then **adding 1**. In short, **two's complement = one's complement + 1**.

- Equivalent rule: starting from the signed binary representation of the positive value, copy bits from right to left unchanged until (and including) the first `1` is copied; then complement (flip) all remaining bits to the left. (An important exception: `10000000` in 8 bits represents -128, which has no positive counterpart in the same width.)
- For an n-bit two's complement number, the representable range is $-(2^{n-1}) \leq D \leq +(2^{n-1}-1)$ – e.g. for 8 bits: -128 to $+127$.
- **Subtraction using two's complement**: simply add the two's complement of the number being subtracted; any carry out of the most significant bit (the 9th bit, for 8-bit numbers) is **simply discarded** (no end-around carry needed). *Example:* $5 + (-3)$ in 8-bit two's complement:

$$00000101 + 11111101 = 1\,00000010 \;\Rightarrow\; \text{discard overflow bit} \;\Rightarrow\; 00000010 = +2 \quad \checkmark$$

- **Benefits of two's complement**: only one representation of zero; arithmetic (addition and subtraction) works uniformly using ordinary binary addition; subtraction $A - B$ can always be computed as $A +$ (two's complement of B).
- **Drawback**: the range is asymmetric (e.g. for 4 bits: $+7$ to -8), and computing the negative of a number requires bit inversion plus an increment (a small amount of extra work compared to just flipping the sign bit).

8-bit two's complement reference table (selected values):

Bit pattern	Value
01111111	+127
⋮	⋮
00000011	+3
00000010	+2
00000001	+1
00000000	0
11111111	−1
11111110	−2
11111101	−3
⋮	⋮
10000001	−127
10000000	−128

2.2.5 Summary of Integer Ranges (8-bit example)

Representation	Largest	Smallest
8-bit unsigned	255	0
8-bit two's complement	127	−128

When subtracting two numbers $a - b$, instead of designing a separate subtraction circuit, it is more practical to **negate b (two's complement) and then add**, since addition hardware (and two's complement negation) is already available – this is exactly why two's complement is the standard representation used by real computer hardware.

2.3 Non-Integer (Fractional) Numbers

2.3.1 Fractions in Decimal and in Binary

A number with a fractional part can be expressed as a sum of place values with negative exponents.

Decimal example:

$$16.357 = 1 \times 10^1 + 6 \times 10^0 + 3 \times 10^{-1} + 5 \times 10^{-2} + 7 \times 10^{-3}$$

Binary example: 10.011_2

$$10.011_2 = 1 \times 2^1 + 0 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3} = 2 + \tfrac{1}{4} + \tfrac{1}{8} = 2\tfrac{3}{8}_{10}$$

2.3.2 Converting a Decimal Fraction to Binary

Convert the integer part normally (repeated division by 2). For the fractional part, repeatedly **multiply by 2**; each time, the digit to the left of the new decimal point (0 or 1) is the next binary fraction digit, and the process continues with the remaining fractional part until it becomes 0 (or the desired precision is reached).

Example: Convert 16.375_{10} to binary.

- Integer part: $16_{10} = 10000_2$.
- Fractional part 0.375:

$$0.375 \times 2 = 0.750 \;\Rightarrow\; 0, \quad 0.750 \times 2 = 1.500 \;\Rightarrow\; 1, \quad 0.500 \times 2 = 1.000 \;\Rightarrow\; 1$$

  Reading the generated digits from first to last: $0.375_{10} = 0.011_2$.

Therefore $16.375_{10} = 10000.011_2$.

2.4 Scientific Notation

Decimal scientific notation:

$$135.26 = 1.3526 \times 10^2, \qquad 13{,}526{,}000 = 1.3526 \times 10^7, \qquad 0.0000002452 = 2.452 \times 10^{-7}$$

In 1.3526×10^7: the **mantissa** (significand) is 3526, the **base** is 10, and the **exponent** is 7.

Binary scientific notation: a binary fraction can similarly be normalised, e.g.

$$0.000001001_2 = 1.001_2 \times 2^{-6}$$

by shifting the radix point so that exactly one non-zero digit remains to its left, and adjusting the exponent accordingly.

2.5 IEEE 754 Binary Floating-Point Representation

2.5.1 32-bit Format

A single-precision (32-bit) floating-point number is laid out as:

$$\underbrace{1 \text{ bit}}_{\text{Sign}} \quad \underbrace{8 \text{ bits}}_{\text{Biased Exponent}} \quad \underbrace{23 \text{ bits}}_{\text{Significand (Mantissa)}}$$

- **Sign bit**: 0 = positive, 1 = negative.

- **Biased exponent** (8 bits): rather than using two's complement or a separate sign for the exponent, a fixed **bias** is *added* to the true exponent before storing it. For 32-bit format the bias is 127 ($= 2^{8-1} - 1$); in general the bias equals $2^{k-1} - 1$ where k is the number of exponent bits.

  $$\text{Biased Exponent} = \text{True Exponent} + 127$$

  With an 8-bit field, the *stored* biased exponent value ranges over 0 to 255, but the two extreme patterns (`00000000` and `11111111`) are reserved for special values (see below), so the usable **true** exponent range is -126 to $+127$.

- **Significand / Mantissa** (23 bits): stores the fractional part of a *normalized* number.

2.5.2 Normalization

A number is **normal** (normalized) if the most significant digit of its significand is non-zero. In binary, a normal non-zero number has the form

$$\pm 1.f_1 f_2 f_3 \ldots \times 2^e$$

Because the leading bit of a normalized binary significand is *always* 1, it does not need to be stored explicitly – it is an **implicit bit**, which effectively gives 24 bits of precision from only 23 stored bits.

- **Explicit normalization**: move the radix point to the *left* of the MSB (used when the number looks like 0.00110_2, giving e.g. 0.110×2^{-2}).
- **Implicit normalization**: move the radix point to the *right* of the MSB (used when the number looks like 101.100_2, giving e.g. 1.01100×2^2).

2.5.3 Worked Example: 32-bit Representation of 263.3

$$\begin{aligned} 263_{10} &= 100000111_2 \\ 0.3_{10} &\approx 0.0100110011001101_2 \\ 263.3_{10} &\approx 100000111.0100110011001101_2 \\ &= 1.00000111010011001100 1101 \times 2^8 \quad \text{(normalized)} \end{aligned}$$

- Sign bit = 0 (positive number).
- True exponent = 8 $\Rightarrow$ Biased Exponent = $8 + 127 = 135 = 10000111_2$.
- Mantissa (drop the implicit leading 1, keep 23 bits, truncating/rounding the LSB as needed): `00000111010011001100110`.

Full 32-bit pattern:

$$0 \;\; 10000111 \;\; 00000111010011001100110$$

2.5.4 Special Values, Overflow and Underflow

- An all-zero biased exponent (-127 true, all-0 field) together with an all-zero significand represents ± 0; more generally, an all-zero exponent field with a non-zero significand represents a **denormalized (subnormal)** number, allowing representation of numbers smaller than the smallest normal number, at the cost of reduced precision.
- An all-one biased exponent ($+128$ true, all-1 field) is reserved: combined with a zero significand it represents $\pm\infty$ (e.g. $1/0 = \infty$); combined with a non-zero significand it represents **NaN** (Not a Number), e.g. $\sqrt{-1}$.

- **Floating-point overflow**: occurs when the true exponent needed exceeds the maximum representable value (127 for 32-bit format); the result effectively becomes $\pm\infty$. *Example:* $1.111111 \times 2^{127} + 1.111111 \times 2^{127} = 11.111110 \times 2^{127} \approx 1.1111110 \times 2^{128}$, which cannot be represented since $128 > 127$.
- **Floating-point underflow**: occurs when the true exponent needed is smaller than the minimum representable value (-126 for 32-bit format); the result effectively becomes 0. *Example:* $10.010000 \times 2^{-126} \div 11.000000 \times 2^{0} = 0.110000 \times 2^{-126} \approx 1.100000 \times 2^{-127}$, which underflows to approximately 0.
- **Range summary (32-bit)**: negative numbers between $-(2 - 2^{-23}) \times 2^{128}$ and -2^{-127}; positive numbers between 2^{-127} and $(2 - 2^{-23}) \times 2^{128}$. The largest finite magnitude is approximately $(2 - 2^{-23}) \times 2^{127} \approx 3.4028235 \times 10^{38}$.

2.5.5 Single vs Double Precision

Format	Total bits	Exponent bits	Significand bits
Single-precision (`float` in C/C++)	32	8	23
Double-precision (`double` in C/C++)	64	11	52

IEEE Standard 754 (adopted 1985, revised 2008) is the most important floating-point standard, developed to make numerical programs portable across processors; it uses this same biased ("excess-N") exponent scheme for various precisions (e.g. excess-15 for half precision, excess-127 for single, excess-1023 for double, excess-16383 for quad precision).

2.6 Limitations of Floating-Point and Rounding Error

- Floating-point representation can only **approximate** real numbers; more bits reduce the error but can never eliminate it entirely.
- Overflow and underflow can cause a program to crash, or silently produce erroneous, hard-to-detect results.
- When a smaller floating-point number's exponent is adjusted to match a larger number before addition, some of its least-significant bits may be lost – this loss of precision is called **rounding**.
- A **round-off (rounding) error** is the difference between the exact mathematical result and the result obtained using finite-precision arithmetic. This is a form of **quantization error**, distinct from a **truncation error** (simply chopping off digits without rounding).
- Two common rounding rules:
  - **Round-by-chop**: if the discarded portion is less than 0.5 (in the units of the last kept digit), the number is simply truncated (chopped).
  - **Round-to-nearest** (used by the IEEE standard): if the discarded portion is greater than 0.5, add 1 to the least significant bit (LSB) of the mantissa, carrying if necessary. If the discarded portion is exactly 0.5, add 1 to the LSB only if the LSB is currently 1 (i.e. round to the nearest *even* value); otherwise leave it unchanged.

  Example (6 significant decimal digits): representing $\pi = 3.14159265358979\ldots$: round-by-chop gives 3.14159; round-to-nearest gives 3.14159 as well here, but for $3.1415935\ldots$ chop would give 3.14159 while round-to-nearest would give 3.14160.

- **Worked example – 16-bit float**: represent 1008.8125_{10} using a 16-bit format with 1 sign bit, 5 exponent bits and 10 mantissa bits.

$$1008_{10} = 1111110000_2, \qquad 0.8125_{10} = 0.1101_2$$

$$1008.8125_{10} = 1111110000.1101_2 = 1.1111100001101 \times 2^9$$

Biased exponent $= 9 + 15 = 24 = 11000_2$ (bias for a 5-bit exponent field is $2^{5-1} - 1 = 15$). Before rounding: `0 11000 1111100001101`. Since the mantissa has more than 10 bits available, it must be rounded to 10 bits, giving: `0 11000 1111100010` (after rounding up the final kept bit).

- **Real-world consequence**: the 1991 Patriot missile failure during the Gulf War (Dhahran, Saudi Arabia) was traced to a computer arithmetic rounding error that caused an inaccurate calculation of elapsed time since system boot, which caused the system to fail to intercept an incoming Scud missile.

3 Boolean Algebra and Logic Operators

3.1 Basics of Boolean Algebra

Boolean algebra was introduced by George Boole (a British mathematician) in 1854. It is the branch of algebra in which variables take only two truth values: **true** and **false**, conventionally written as 1 and 0.

The three basic Boolean operations are:

Operation	Symbol(s)	Name
AND	$A \cdot B$ or $A \wedge B$	Conjunction
OR	$A + B$ or $A \vee B$	Disjunction
NOT	$\overline{A}$ or $\neg A$	Negation

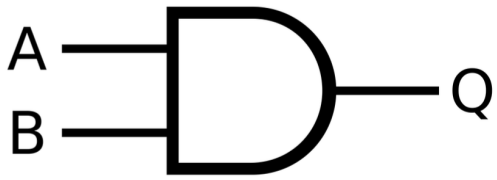
A **Boolean function** is an expression built from binary variables combined using Boolean operators, e.g. $F(x, y) = x.\overline{y} + \overline{x}.(x + y)$. Since each variable can only take 2 values, a function of n variables has 2^n possible input combinations, and can be fully described by a **truth table** with 2^n rows.

3.2 Logic Gates

A **logic gate** is a piece of hardware (built internally from transistors, resistors and diodes) that implements a Boolean operation on its inputs. The behavioural diagram of any Boolean expression can be represented as a circuit built from logic gates.

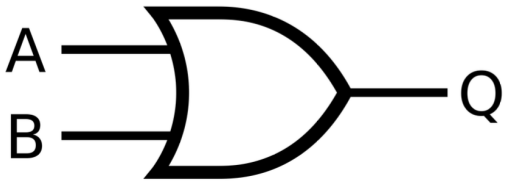
3.2.1 AND Gate

$Q = A \cdot B$



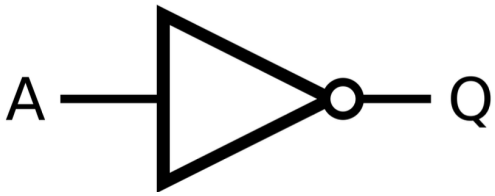
3.2.2 OR Gate

$Q = A + B$



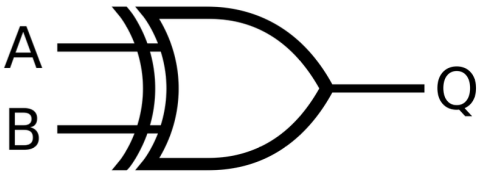
3.2.3 NOT Gate

$Q = \overline{A}$



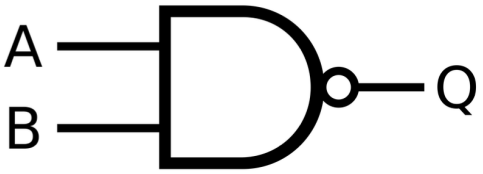
3.2.4 XOR Gate

$Q = A \oplus B$ (output is 1 only when the inputs differ)



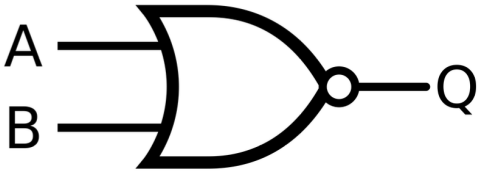
3.2.5 NAND Gate

$Q = \overline{A \cdot B}$



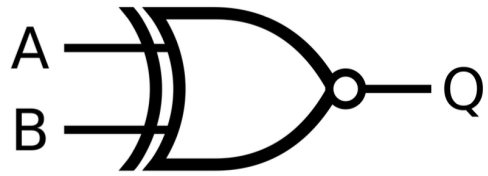
3.2.6 NOR Gate

$Q = \overline{A + B}$



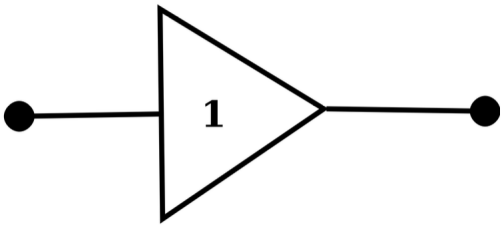
3.2.7 XNOR Gate

$Q = \overline{A \oplus B}$ (output is 1 only when the inputs are equal)



3.2.8 Buffer

A buffer passes its input to its output unchanged ($Q = A$). It is used to increase the propagation delay of a circuit, e.g. to help match timing between paths.



3.2.9 Multiple-Input Gates

Gates with more than two inputs are also available (typically built into ICs, one gate type per IC, sometimes with multiple gates per chip). For example:

$F = A \wedge B \wedge C$ (3-input AND), $F = A \vee B \vee C$ (3-input OR), $F = A \oplus B \oplus C$ (3-input XOR)

3.2.10 Operator Precedence

When evaluating or simplifying Boolean expressions, the operator precedence (highest to lowest) is:

$$() \ > \ \text{NOT}(\sim) \ > \ \text{AND}(\wedge) \ > \ \text{OR}(\vee)$$

3.3 Drawing Logic Circuits from Expressions

Any Boolean expression can be translated directly into a circuit diagram by substituting each operator for its corresponding gate and wiring the variables into the correct gates, respecting operator precedence. More complex expressions require more gates wired together; note that the *same* Boolean function can often be drawn as several different circuit layouts (different gate arrangements) that are logically equivalent but not necessarily equally efficient.

3.4 Truth Tables and Boolean Expressions

3.4.1 From an Expression to a Truth Table

Given a Boolean expression with n variables, there are 2^n possible input combinations, so the truth table will have 2^n rows. Substitute each combination of variable values into the expression to compute the corresponding output.

Example: $F(x, y, z) = x.y.\overline{z} + y.\overline{z}$ has 3 variables, so its truth table has $2^3 = 8$ rows, computed by substituting each of the 8 combinations of x, y, z.

3.4.2 Canonical Forms: Minterms and Maxterms

When converting a truth table into a Boolean expression, there are two standard *canonical* forms, in which every term must contain **all** the variables of the function.

- **Minterm (Standard Product)**: a term formed by AND-ing together all the variables of the function, each either in its normal or complemented form. For 3 variables there are $2^3 = 8$ possible minterms (e.g. $\overline{x}.\overline{y}.\overline{z}$, $\overline{x}.\overline{y}.z$, ..., $x.y.z$). Note that a term like $(x.y)$ alone is *not* a minterm for a 3-variable function, since it is missing z.
- **Maxterm (Standard Sum)**: a term formed by OR-ing together all the variables of the function, each either in its normal or complemented form. For 3 variables there are $2^3 = 8$ possible maxterms (e.g. $\overline{x} + y + z$, ..., $x + y + z$).
- **Sum of Minterms (SoM)**: a Boolean function expressed as an OR (disjunction) of minterms – also called **Sum of Products** in canonical form.
- **Product of Maxterms (PoM)**: a Boolean function expressed as an AND (conjunction) of maxterms – also called **Product of Sums** in canonical form.

Standard forms (as opposed to canonical forms) allow terms that do *not* need to contain every variable of the function, e.g. $F = \overline{x}.y + x$ is in **Sum of Products** standard form even though the term x alone does not mention y.

3.4.3 Converting a Truth Table to Sum of Minterms

Every row of the truth table where the output is 1 corresponds to exactly one minterm: for that row, write each variable un-complemented if it is 1 in that row, and complemented if it is 0. The overall function is the OR (disjunction) of the minterms for every row where the output is 1 (this works because the expression can only equal 1 when the input matches one of these specific rows – a disjunction of the conditions under which the output is true).

Example: for the rows (indexed from 0) where the output is 1: 2, 4, 5, 6, 7, we would write $F(x, y, z) = \sum m(2, 4, 5, 6, 7)$ using **minterm shorthand notation**, meaning the sum (OR) of minterms m_2, m_4, m_5, m_6, m_7.

3.4.4 Converting a Truth Table to Product of Maxterms

Symmetrically: every row where the output is 0 corresponds to one maxterm. For that row, write each variable complemented if it is 1 in that row, and un-complemented if it is 0 (i.e. the opposite convention from minterms), then AND all these maxterms together.

This can be justified via **De Morgan's Theorem**: if F' (the complement of F) can be written as a sum of minterms (using the rows where $F = 0$), then $F = \overline{F'}$ can be obtained by applying De Morgan's theorem to convert that sum of (complemented) minterms into a product of maxterms.

Shorthand notation: $F(x, y, z) = \prod M(0, 1, 3)$ means the product (AND) of maxterms M_0, M_1, M_3 (i.e. rows 0, 1 and 3 have output 0).

3.4.5 De Morgan's Theorem

$$\overline{A \cdot B} = \overline{A} + \overline{B}, \qquad \overline{A + B} = \overline{A} \cdot \overline{B}$$

This theorem generalises to any number of variables, and is essential for converting between Sum-of-Products and Product-of-Sums forms, and for implementing circuits using a single gate type (see Section 3.8).

3.4.6 Converting a Canonical Form Back to a Truth Table

- **From SoM**: for each minterm in the expression, find the truth table row it corresponds to and fill in a 1 in the output column; fill all remaining rows with 0.

- **From PoM**: for each maxterm, negate every literal in it and AND them together to find the corresponding row (by De Morgan's theorem, this identifies the row that makes the *original* maxterm equal to 0); fill in a 0 for that row. Fill all remaining rows with 1.

3.5 Boolean Algebraic Laws and Theorems

Law		
Annulment Law	$A \cdot 0 = 0$	$A + 1 = 1$
Identity Law	$A + 0 = A$	$A \cdot 1 = A$
Idempotent Law	$A + A = A$	$A \cdot A = A$
Complement Law	$A + \overline{A} = 1$	$A \cdot \overline{A} = 0$
Commutative Law	$A \cdot B = B \cdot A$	$A + B = B + A$
Double Negation Law	$\overline{\overline{A}} = A$	
Distributive Law	$A(B + C) = A.B + A.C$	$A + (B.C) = (A + B).(A + C)$
Redundancy Law	$A + A.B = A$	$A.(A + B) = A$
De Morgan's Law	$\overline{A.B} = \overline{A} + \overline{B}$	$\overline{A + B} = \overline{A}.\overline{B}$

There are also two additional useful identities sometimes listed as "absorption-type" laws:

$$A + \overline{A}.B = A + B, \qquad A.(\overline{A} + B) = A.B$$

3.6 Simplifying Boolean Expressions Algebraically

A Boolean function can always be expressed in canonical SoM or PoM form, but such expressions are rarely the simplest possible. Simpler expressions require fewer gates, which is cheaper and more efficient to manufacture, so simplification (using the laws above, applied step by step) is an important skill.

Worked example: Simplify $F = A + B.C + \overline{C}.(A + B.D)$.

$$\begin{aligned} F &= A.A + A.B.C + A.\overline{C} + A.B.D + B.C.\overline{C} + B.D.\overline{C} \\ &= A(1 + B.C + \overline{C} + B.D) + B.C.\overline{C} + B.D.\overline{C} \\ &= A + B.D.\overline{C} \end{aligned}$$

(using $A.A = A$, $A{\cdot}1 = A$, and $B.C.\overline{C} = 0$ from the Complement Law, then absorbing remaining terms into A via the Redundancy Law).

De Morgan's theorem is frequently combined with the other laws when simplifying expressions that contain complemented sums or products, e.g. $\overline{A + B}$ must first be rewritten as $\overline{A}.\overline{B}$ before further simplification can proceed algebraically.

3.7 Karnaugh Maps (K-Maps)

A **Karnaugh map** is a graphical, gate-level minimization technique: it represents a truth table as a grid of cells, where each cell corresponds to exactly one minterm, arranged so that any two *physically adjacent* cells (horizontally or vertically, including wrap-around between opposite edges of the map) differ in exactly one variable. If the well-defined grouping rules below are followed correctly, a K-map is guaranteed to produce the simplest possible Sum-of-Products (or Product-of-Sums) expression.

For a function of n variables, the K-map has 2^n cells (e.g. a 3-variable K-map has 8 cells, typically arranged as a 2×4 grid).

3.7.1 K-Map Grouping Rules

1. **Single type only**: a group must contain either all 1s (if deriving a Sum-of-Products expression) or all 0s (if deriving a Product-of-Sums expression) – never a mix.
2. **Rectangular, not diagonal**: groups must be formed by expanding horizontally or vertically; diagonal groupings are not allowed.
3. **Size must be a power of two**: a group may only contain 2^n cells (i.e. 1, 2, 4, 8, …cells), for $n \geq 0$.
4. **Maximum size**: every group should be expanded to be as large as possible (subject to the other rules).
   - Groups *may overlap* with each other if doing so allows a group to reach its maximum possible size.
   - The map **wraps around**: the top row is considered adjacent to the bottom row, and the leftmost column is considered adjacent to the rightmost column, so groups may be formed across these edges.
5. **Fewest groups**: use as few groups as possible while still covering every required cell and keeping every group at its maximum size.

Grouping 2^k cells together eliminates exactly k variables from the resulting term, because those k variables take on every possible combination of values within the group while the remaining (unchanging) variables define the simplified term for that group. This is precisely why groups must have size 2^k and be formed only along rows/columns (never diagonally) – diagonal or non-power-of-two groupings would not correspond to eliminating a whole variable cleanly across all its possible values.

3.7.2 Don't Care Conditions

Sometimes a function is **incompletely specified**: certain input combinations either cannot occur in practice, or their output value simply does not matter. These input patterns are called **don't care conditions**, marked with an 'X' (or 'd') in the K-map. When grouping, an 'X' can be treated as either a 0 or a 1 – whichever choice produces the largest, most efficient groupings – since its actual value does not affect correctness.

3.7.3 Sum of Products vs Product of Sums from a K-Map

- To derive a simplified **Sum of Products** expression: group the **1s**.
- To derive a simplified **Product of Sums** expression: group the **0s**, then apply De Morgan's theorem appropriately to each group (each group of 0s gives one *sum* term in the final product).

3.8 Functional Completeness and Universal Gates

AND, OR and NOT are considered the three "primary" Boolean functionalities, since every possible Boolean function can be built from some combination of them. A set of gates that can, between them, implement all three of AND, OR and NOT is called a **functionally complete set**. By definition, {AND, OR, NOT} is functionally complete.

Other functionally complete sets include:

- {AND, NOT} – since $A + B = \overline{\overline{A}.\overline{B}}$ (OR can be built from AND and NOT, via De Morgan's theorem).

- {OR, NOT} – since $A.B = \overline{\overline{A} + \overline{B}}$ (AND can be built from OR and NOT, via De Morgan's theorem).
- {NAND} alone – NAND is **individually** functionally complete.
- {NOR} alone – NOR is **individually** functionally complete.

Because NAND and NOR are each individually functionally complete, they are called **Universal Gates**: any Boolean function whatsoever can be implemented using *only* NAND gates, or *only* NOR gates. This matters practically because manufacturing an IC containing only one type of gate is cheaper and simpler than manufacturing ICs with a mixture of gate types (which can also leave unused gates wasted on the chip).

Building NOT, AND, OR from NAND alone:

- NOT: tie both inputs of a NAND gate together: $\overline{A.A} = \overline{A}$.
- AND: a NAND gate followed by a NAND-based NOT (inverter): $\overline{\overline{A.B}} = A.B$.
- OR: invert both inputs first (using NAND-based NOT), then feed them into a NAND gate: $\overline{\overline{A}.\overline{B}} = A + B$ (by De Morgan's theorem).

Building NOT, AND, OR from NOR alone:

- NOT: tie both inputs of a NOR gate together: $\overline{A + A} = \overline{A}$.
- AND: invert both inputs first (using NOR-based NOT), then feed them into a NOR gate: $\overline{\overline{A} + \overline{B}} = A.B$ (by De Morgan's theorem).
- OR: a NOR gate followed by a NOR-based NOT (inverter): $\overline{\overline{A + B}} = A + B$.

When converting a circuit built from mixed gate types into an all-NAND (or all-NOR) circuit by direct substitution, any place where two inversions cancel out (double negation) can be removed to simplify the resulting circuit.

4 Logic Circuits

Logic circuits are broadly classified into two categories:

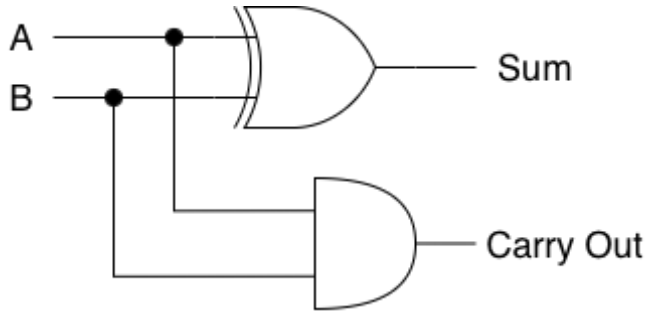
- **Combinational logic circuits**: the output depends only on the current values of the inputs; there is no memory of past inputs.
- **Sequential logic circuits**: the output depends on both the current inputs and the circuit's previous state, which is held in memory elements.

4.1 Combinational Circuit Design

4.1.1 Adders

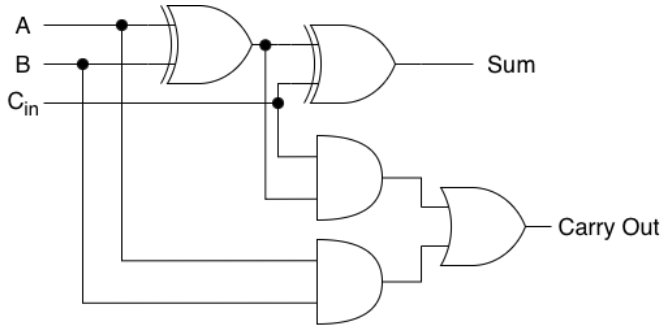
Half Adder:

- Adds two single bits, A and B.
- Cannot accept a carry-in from a previous stage, so it produces two outputs: Sum and Carry Out.
- Sum $= A \oplus B$, Carry Out $= A.B$



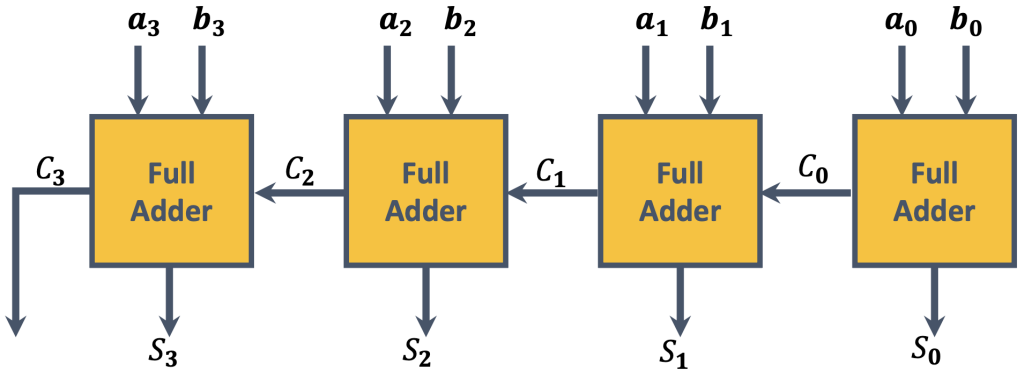
Full Adder:

- Adds three bits: A, B, and a carry-in C_{in} (the carry produced by the previous less-significant stage).
- Can be built by combining two half adders and an OR gate, or directly from AND/OR/XOR gates as shown below.
- Sum $= A \oplus B \oplus C_{in}$
- Carry Out $= A.B + A.C_{in} + B.C_{in}$



Ripple Carry Adder:

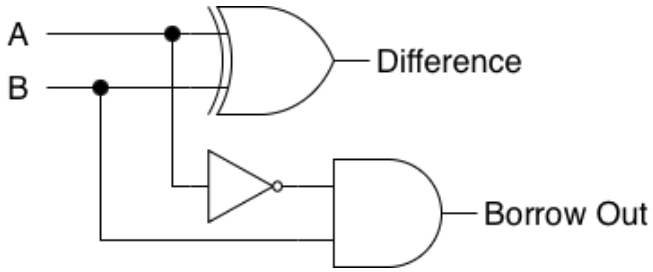
- Chains multiple full adders together to add multi-bit binary numbers.
- For n-bit numbers, n full adders are required, with the carry-out of each stage feeding into the carry-in of the next more-significant stage.
- Called "ripple carry" because the carry signal must propagate (ripple) through every stage in sequence, which limits the speed of the adder for large bit-widths.



4.1.2 Subtractors

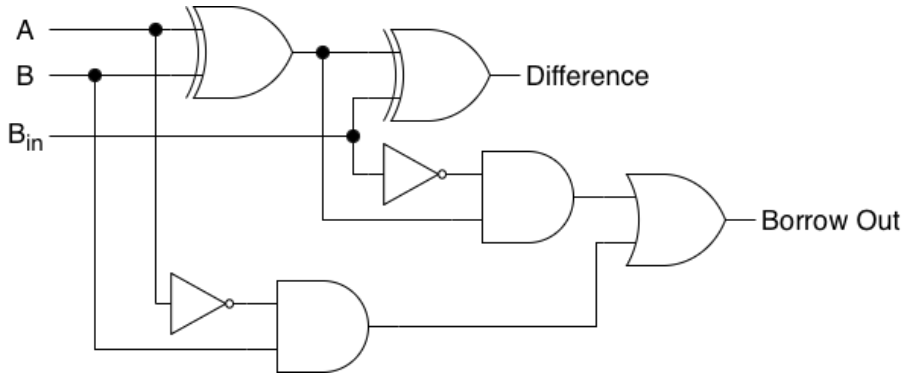
Half Subtractor:

- Subtracts one bit B from one bit A, producing a Difference and a Borrow Out.
- Cannot accept a borrow-in from a previous stage.
- Difference $= A \oplus B$, Borrow Out $= \overline{A}.B$



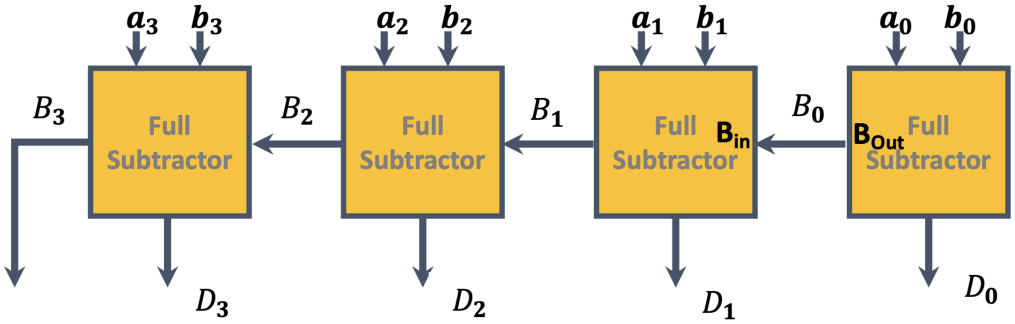
Full Subtractor:

- Subtracts three bits: A, B, and a borrow-in B_{in}.
- Can be built from two half subtractors and an OR gate.
- Difference $= A \oplus B \oplus B_{in}$
- Borrow Out $= \overline{A}.B + \overline{A}.B_{in} + B.B_{in}$



Ripple Borrow Subtractor:

- Chains multiple full subtractors together to subtract multi-bit binary numbers.
- For n-bit numbers, n full subtractors are required, with the borrow-out of each stage feeding into the borrow-in of the next more-significant stage.



4.1.3 Multipliers

Binary multiplication of two bits mirrors long multiplication:

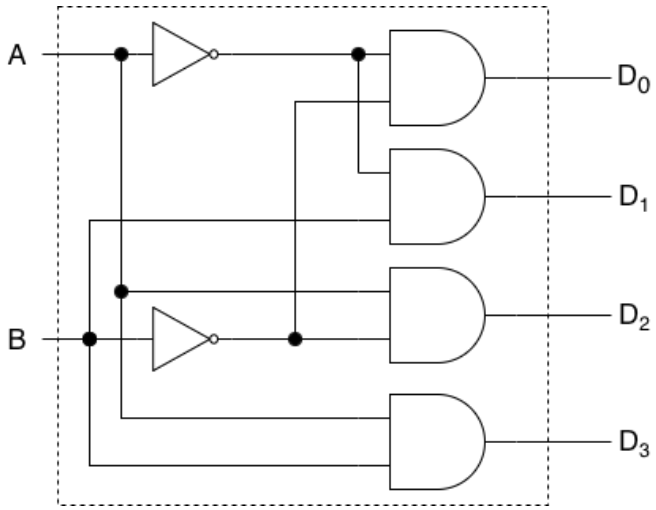
- **Generation of partial products**: each bit of one operand multiplies (logically ANDs with) every bit of the other operand.
- **Summation of partial products**: the partial products are added together, each one shifted one bit position further to the left than the previous.
- In hardware: AND gates generate each partial product bit, and a network of full adders (arranged similarly to a ripple carry adder) sums the shifted partial products; the physical wiring pattern implements the required left-shifting.

4.1.4 Decoders

- A **decoder** interprets an encoded input (a code word) and activates exactly one of several output lines corresponding to that code.
- For n input lines, a decoder can support up to 2^n unique output lines (e.g. 2 inputs and $2^2 = 4$ outputs is called a 2×4 decoder).
- Most decoders include an **Enable (E)** input; when Enable is 0 the entire circuit is disabled and all outputs remain 0 regardless of the other inputs.
- Decoders are useful **circuit builders**, because each output line of an n-to-2^n decoder is exactly one minterm of the n inputs. Any Boolean function expressed in Sum-of-Minterms form can therefore be built by feeding the appropriate decoder outputs into an external OR gate.

2×4 **Decoder** (inputs A, B; outputs D_0–D_3):

$$D_0 = \overline{A}.\overline{B}, \quad D_1 = \overline{A}.B, \quad D_2 = A.\overline{B}, \quad D_3 = A.B$$

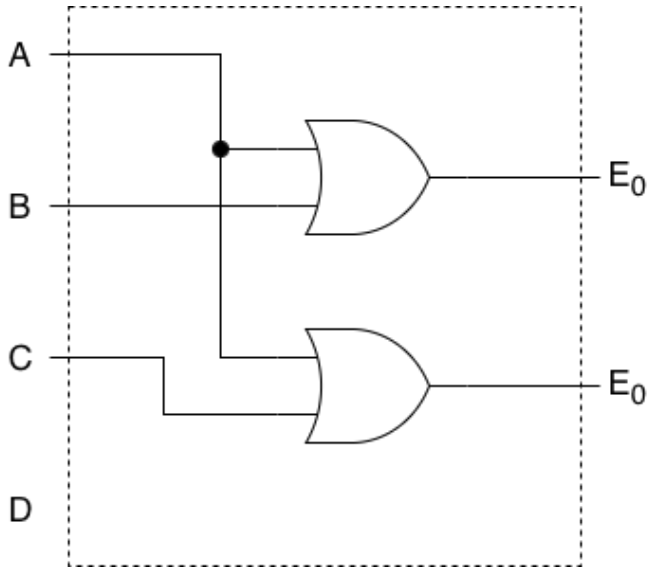


4.1.5 Encoders

- An **encoder** performs the reverse operation of a decoder: it converts an "active" input line back into its corresponding binary code.
- For n output channels, an encoder requires 2^n input channels (e.g. 4 inputs and 2 outputs is called a 4×2 encoder).
- A basic encoder normally assumes exactly one input is active at a time; if more than one input can be active simultaneously, a **priority encoder** is needed to resolve which input takes precedence.

4×2 **Encoder** (inputs A, B, C, D; outputs E_1, E_0):

$$E_0 = A + B, \qquad E_1 = A + C$$

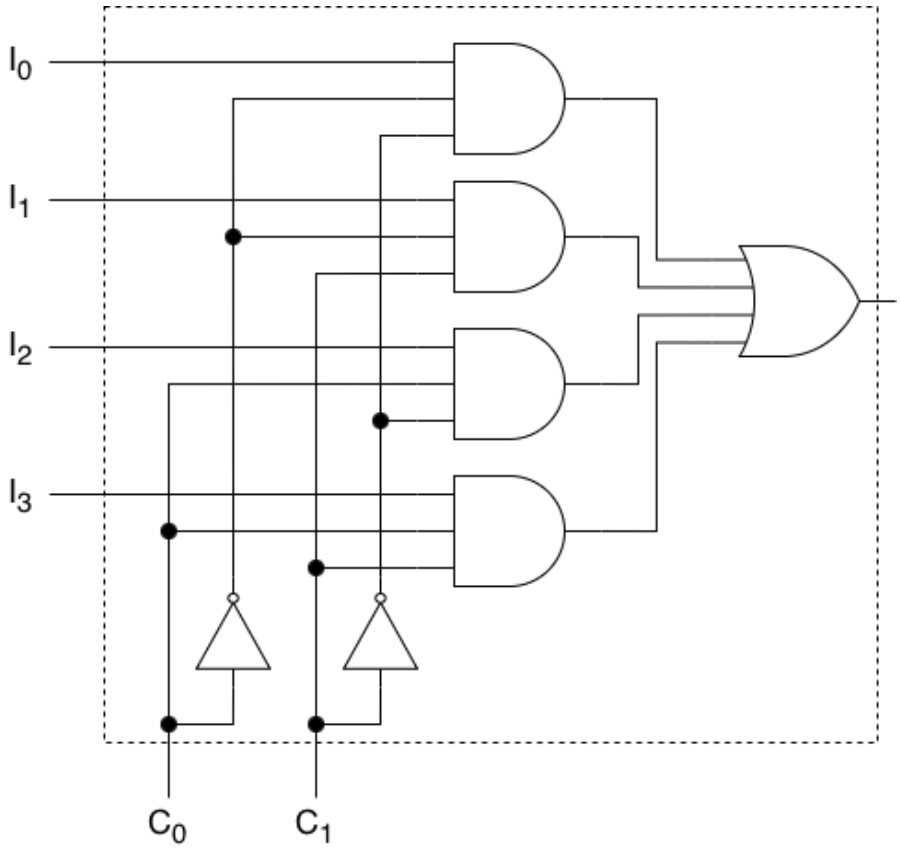


4.1.6 Multiplexers (MUX)

- A **multiplexer** is a digital switch: it has several data input lines but only a *single* output line.
- Its purpose is to select exactly one of the input signals and forward it (route it) to the output.
- Selection is performed using dedicated **control (select) lines**; each unique combination of control-line values acts like a "passcode" that selects one specific input.
- For 2^n data inputs, n control lines are required.

4×1 **Multiplexer** (inputs I_0–I_3; control lines C_0, C_1):

Input	Control Code		Input Activation Condition
	C_0	C_1	
I_0	0	0	$I_0.\overline{C_0}.\overline{C_1}$
I_1	0	1	$I_1.\overline{C_0}.C_1$
I_2	1	0	$I_2.C_0.\overline{C_1}$
I_3	1	1	$I_3.C_0.C_1$



The overall output is the OR of all four activation terms, so only the selected input's value can reach the output at any given time (all other AND gates output 0 because their control-line condition is not satisfied).

4.2 Sequential Logic Circuits

- In a sequential circuit, the output depends on both the present inputs *and* the previous outputs, which are held in a **memory element**.
- The memory element's transition between states is governed by its current state and its inputs; a circuit can therefore be viewed as a time sequence of inputs, outputs and internal states.

4.2.1 Synchronous vs Asynchronous Sequential Circuits

Synchronous Sequential Circuits:

- State changes occur only at discrete time intervals, governed by a global **clock** signal.
- All memory elements update simultaneously, on either the rising or falling edge of the clock pulse.
- *Advantages*: easy to design since all components change state together; hazards and race conditions are minimised since every operation must wait for a clock edge; easy to build complex circuits (e.g. CPUs) because the timing constraint is global and uniform.
- *Disadvantages*: overall speed is limited by the *slowest* path in the circuit; higher power consumption from distributing the clock signal everywhere; in large systems it becomes difficult to ensure the clock reaches every component at exactly the same instant (clock skew).

Asynchronous Sequential Circuits:

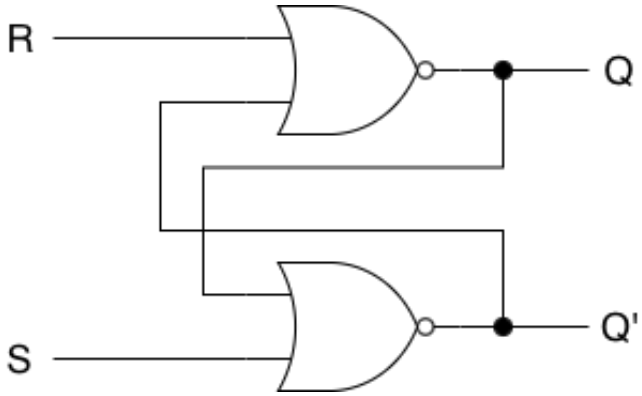
- State changes occur immediately in response to input changes; there is no global clock.
- *Advantages*: faster, since (without a global clock constraint) the circuit can operate at the maximum speed allowed by its individual components; lower power consumption, since only the components actually involved in an operation consume energy; the problem of synchronising a clock signal across the whole circuit is eliminated entirely.
- *Disadvantages*: harder to design correctly; circuits may become unstable since different signals can arrive at different times (leading to glitches/hazards).

4.2.2 Latches

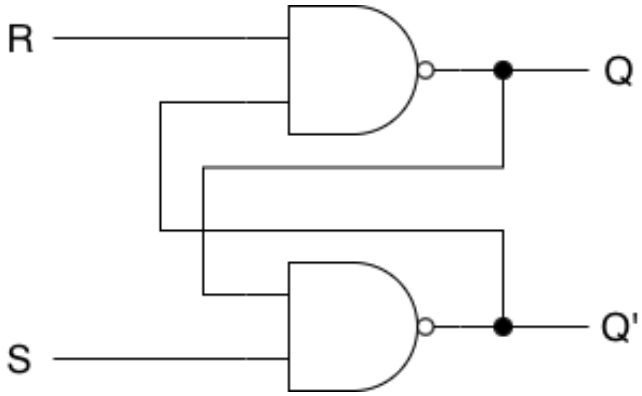
A **latch** is **level sensitive** and **transparent**: as long as its enable signal is active (high), the output continuously tracks the input, and changes immediately whenever the input changes. Latches are asynchronous devices (they do not operate on clock edges), and each is able to store one bit.

S-R Latch (built from NOR gates): the most basic memory building block, with two inputs, Set (S) and Reset (R), and two (complementary) outputs Q and Q'.

S	R	Q	
0	0	Q	No Change
0	1	1	Set
1	0	0	Reset
1	1	0*	Forbidden

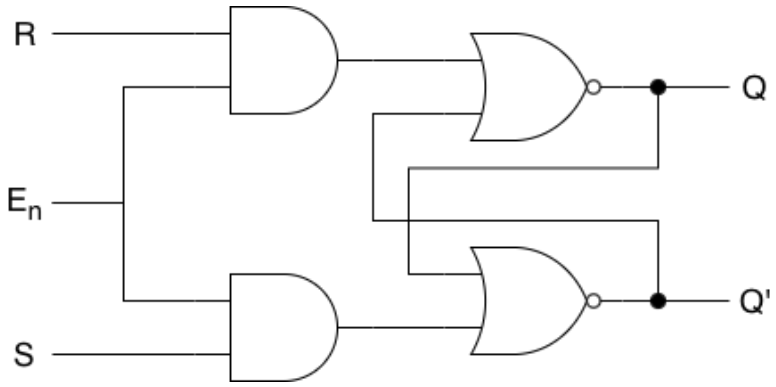


S-R Latch (built from NAND gates), an equivalent circuit using active-low inputs, with its own (dual/inverted) truth-table behaviour:



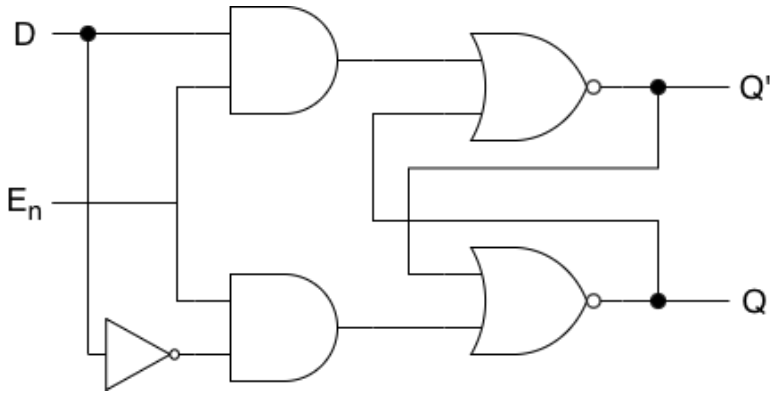
The $S = R = 1$ (NOR version) / $S = R = 0$ (NAND version) case is **forbidden**: it forces Q and Q' to both be equal (violating the requirement that they be complementary), and the resulting next state becomes unpredictable (a race condition) when both inputs are released simultaneously.

S-R Latch with Enable (E_n): adds a control (enable) input so that the latch only responds to S and R while E_n is active (asserted); when E_n is 0 the latch holds its current state regardless of S and R.



D Latch: improves on the S-R latch by eliminating the forbidden-state problem. It has a single Data input (D) and an Enable input (E_n); because D is fed (in complemented and uncomplemented form) into what were the S and R inputs, S and R can never both be asserted at once, so the forbidden state cannot occur.

E_n	D	Q	
0	X	Q	No Change
1	0	0	
1	1	1	



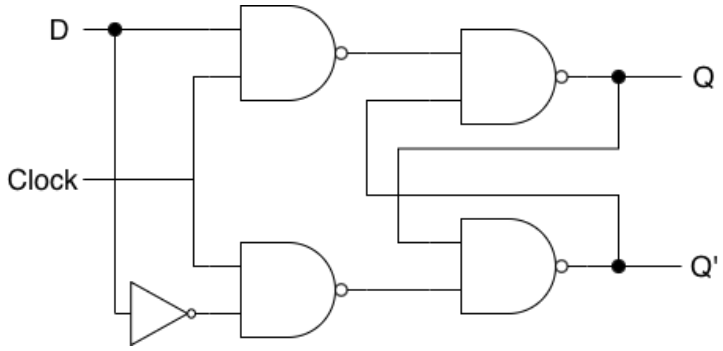
The value of D only reaches the output while the circuit is enabled; when disabled, the output remains unchanged (holds its last value).

4.2.3 Flip-Flops

A **flip-flop** is designed to be more stable than a latch, because it is **edge-triggered** rather than level-sensitive: instead of responding continuously whenever the clock/enable signal is high, it responds only at the instant of a clock **transition** (edge) – either the **positive edge** (rising, $0 \rightarrow 1$) or the **negative edge** (falling, $1 \rightarrow 0$). This edge sensitivity solves the problem where a level-sensitive latch could produce multiple output changes during a single active clock pulse if its input changed while enabled. Like a latch, a flip-flop stores one bit.

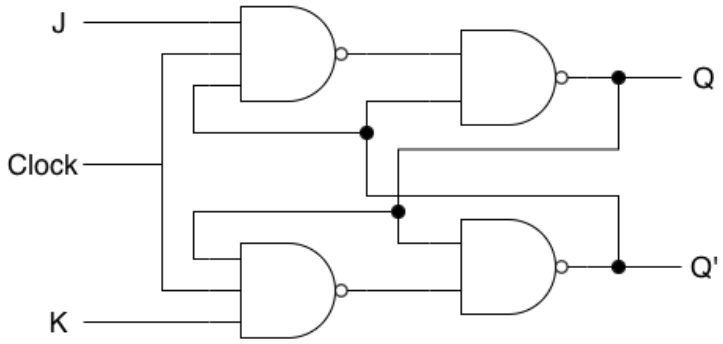
D Flip-Flop: the most widely used flip-flop in digital systems. It acts like a controlled buffer: it captures whatever value is present on the D (Data) input and transfers it to output Q at the next active clock edge.

D	Q	
0	0	Reset
1	1	Set



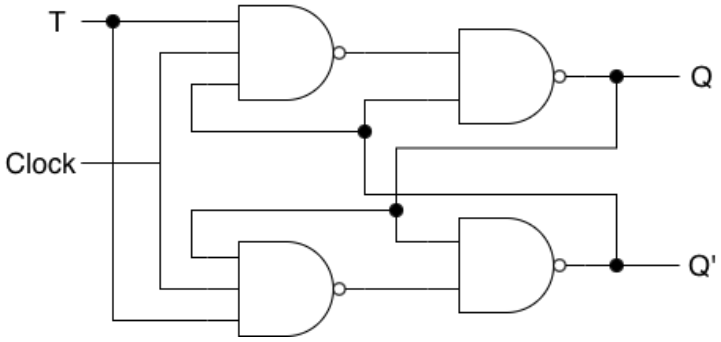
J-K Flip-Flop: an enhanced version of the S-R latch concept, adding a feedback mechanism from the outputs back to the input gates so that the previously-forbidden input combination is instead repurposed to make the flip-flop **toggle** its state.

J	K	Q	
0	0	Q	No Change
0	1	0	Reset
1	0	1	Set
1	1	$\overline{Q}$	Toggle



T Flip-Flop: a simplified special case of the J-K flip-flop in which the J and K inputs are tied together and driven by a single input, T (Toggle).

T	Q	
0	Q	No Change
1	$\overline{Q}$	Toggle



Because it toggles on every active clock edge when $T = 1$, the T flip-flop is especially useful for building binary counters.

4.2.4 Registers

A **register** is a group of flip-flops (typically D flip-flops), one per bit, sharing a common clock, used to store a multi-bit value (e.g. an 8-bit register holds one byte, built from 8 D flip-flops).

4.2.5 Counters

A **counter** is a register (built from flip-flops, most commonly T or J-K flip-flops used in toggle mode) whose state cycles through a fixed sequence of values – typically the binary count sequence – once per active clock edge/toggle condition. Counters are broadly classified into **Ripple (Asynchronous) Counters** and **Synchronous Counters**.

4.2.6 Ripple (Asynchronous) Counters

In a **ripple counter**, the flip-flops are *not* all driven by the same clock signal. Instead, only the *first* flip-flop (representing the least significant bit) is connected to the actual external clock; every subsequent flip-flop is clocked by the **output** of the *previous* flip-flop. Because each flip-flop's output only changes some propagation delay *after* its own clock input changes, and that output is what clocks the next flip-flop, a change at the input takes time to "ripple" through every stage in turn – this is exactly where the name comes from, and it is directly analogous to how carry propagates through a ripple carry adder.

Construction of a 4-bit ripple up-counter using T flip-flops:

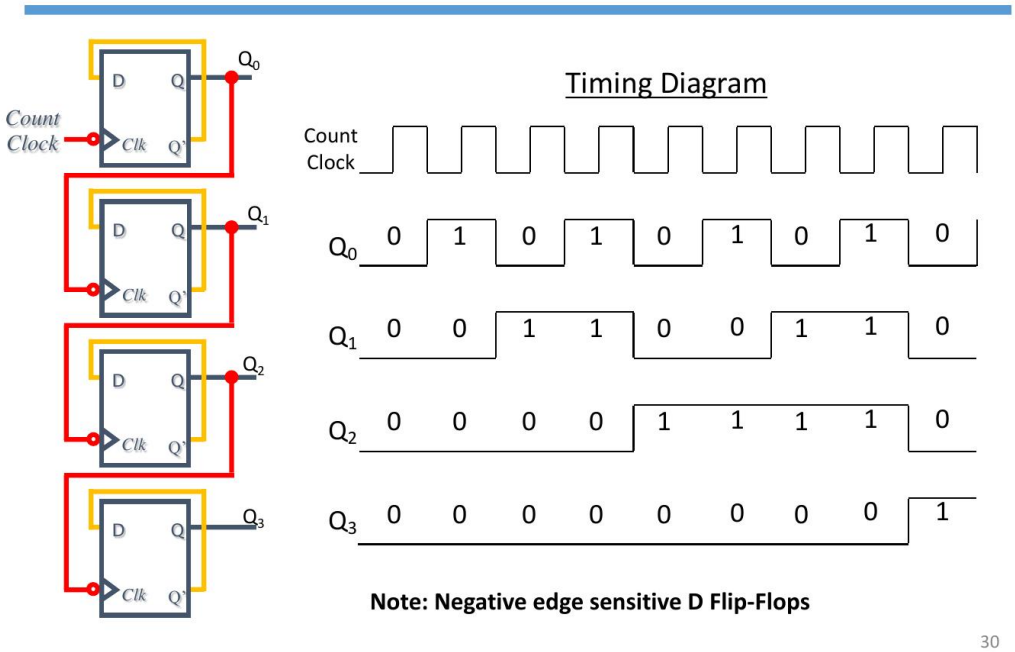
- Each flip-flop is wired in **toggle mode** (for a T flip-flop, this simply means tying $T = 1$ permanently; for a J-K flip-flop, tying $J = K = 1$ permanently), so that every flip-flop toggles its output ($Q \to \overline{Q}$) on every active edge that reaches its clock input.
- The external clock signal is connected only to the clock input of the **first** (least significant bit, Q_0) flip-flop.
- The Q output of each flip-flop is connected to the clock input of the *next* flip-flop in the chain: $Q_0 \to$ Clock of $Q_1 \to$ Clock of $Q_2 \to$ Clock of Q_3, and so on for as many bits as required.
- Whether the counter triggers on the rising edge or the falling edge of each Q output (i.e. whether Q or $\overline{Q}$ feeds the next stage's clock) determines whether the counter counts **up** or **down**; using the normal Q output with negative-edge-triggered flip-flops (or $\overline{Q}$ with positive-edge-triggered flip-flops) produces an up-counter, while the opposite wiring produces a **ripple down-counter**.

How counting emerges from toggling: Since Q_0 toggles on every single clock pulse, it changes state twice as often as Q_1 (which only toggles when Q_0 falls/rises, i.e. once every *two* clock pulses), which in turn changes twice as often as Q_2 (once every *four* clock pulses), and so on. This is exactly the pattern of a binary counting sequence, where bit i (counting from 0) changes precisely once every 2^{i+1} input pulses – so the collection of flip-flop outputs $Q_3Q_2Q_1Q_0$, read together, steps through the ordinary binary count sequence $0000, 0001, 0010, 0011, \ldots$ as clock pulses are applied.

4-bit ripple up-counter state sequence (as pulses arrive at Q_0's clock input):

Pulse count	Q_3	Q_2	Q_1	Q_0
0 (initial/reset)	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
⋮		⋮		
15	1	1	1	1
16 (overflow, wraps to)	0	0	0	0

Circuit and timing diagram (4-bit binary ripple counter, built from negative-edge D flip-flops with $\overline{Q}$ fed back into D, so each flip-flop toggles every time its clock input sees a falling edge):



Reading the timing diagram confirms the ripple pattern directly: Q_0 toggles on every falling edge of the Count Clock; Q_1 only toggles on a falling edge of Q_0 (i.e. half as often); Q_2 only toggles on a falling edge of Q_1 (a quarter as often); and Q_3 only toggles on a falling edge of Q_2 (an eighth as often) – each stage's clock input is literally the previous stage's output, which is

why the transitions visibly "ripple" one stage at a time from left to right across the diagram, rather than all four outputs changing at exactly the same instant.

An n-bit ripple counter built this way has 2^n distinct states (a **MOD-2^n** counter), counting from 0 to $2^n - 1$ before it **overflows** and wraps back around to 0. For example, a 4-bit ripple counter is a MOD-16 counter.

MOD-N (non-power-of-two) ripple counters: if a counter needs to cycle through fewer than 2^n states (e.g. a MOD-10, or "decade", counter that counts 0 to 9 and then resets), extra combinational logic (typically a NAND gate) is added to detect the first unwanted state (e.g. state 1010 for a MOD-10 counter built from 4 flip-flops) and immediately force **asynchronous reset** (clear) on all flip-flops, so the counter skips directly back to 0000 instead of continuing to count further.

Timing/propagation delay: let t_{pd} be the propagation delay of a single flip-flop (the time between an active clock edge arriving and the output actually changing). In an n-bit ripple counter, the *worst-case* total delay before every bit has settled to its correct final value – which occurs at the transition where every single bit must change, e.g. from 0111 to 1000 – is the *sum* of the individual delays along the whole chain:

$$\text{Worst-case settling delay} = n \times t_{pd}$$

This cumulative delay is the main disadvantage of ripple counters: it limits the maximum usable clock frequency, since the next input pulse must not arrive until the ripple effect from the previous pulse has fully settled through every stage, or the counter can briefly pass through incorrect, invalid intermediate states (sometimes visible as glitches on the outputs).

Advantages of ripple counters:

- Very simple to design and build – minimal extra logic gates are needed beyond the flip-flops themselves (no combinational "next-state" logic is required, unlike synchronous counters).
- Consume relatively little power and use less hardware for a given number of bits.

Disadvantages of ripple counters:

- **Cumulative propagation delay**: as shown above, the total worst-case delay grows linearly with the number of bits ($n \times t_{pd}$), which limits the maximum operating (clock) frequency achievable, especially for counters with many bits.
- **Momentary invalid/glitch states**: because the flip-flops do not all change at exactly the same instant, the counter's output can briefly show an incorrect intermediate binary value while the change is still rippling through – this can cause problems if the counter's output is being decoded or acted upon by other circuitry during that brief window.
- Not suitable for high-speed counting applications, where **synchronous counters** (all flip-flops sharing one common clock, so every bit changes simultaneously) are preferred instead, at the cost of requiring extra combinational logic to compute each flip-flop's correct next-state input.

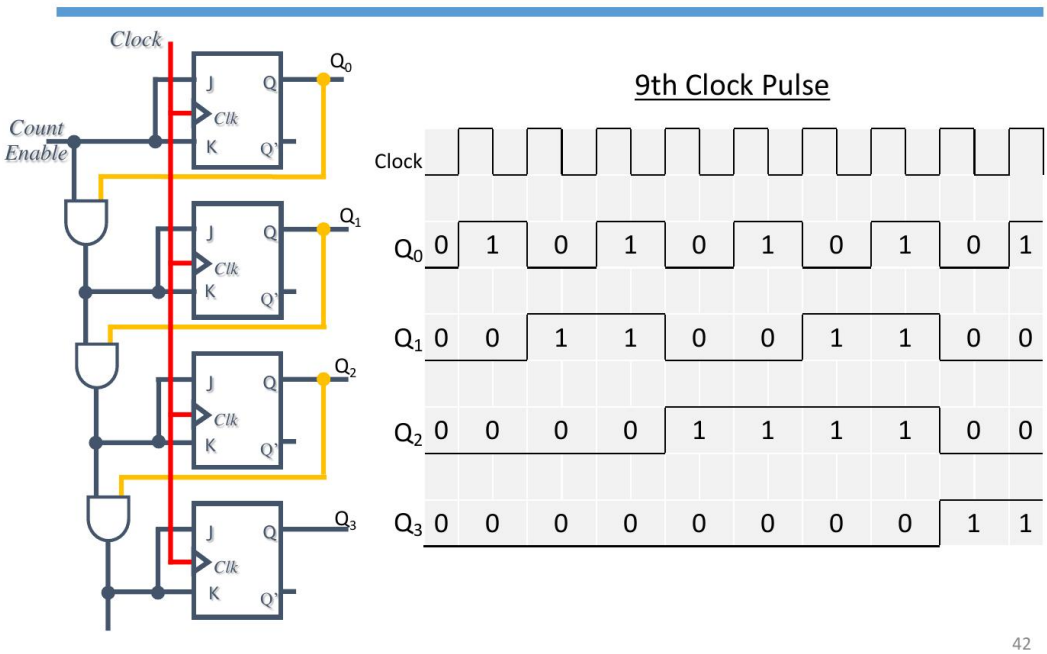
4.2.7 Synchronous Counters

In a **synchronous counter**, every flip-flop is triggered by the *same, common* clock signal, rather than by the output of the previous flip-flop. Since all flip-flops receive the active clock edge at exactly the same instant, every bit that needs to change *does* change simultaneously – there is no ripple delay, and no possibility of the counter briefly showing an incorrect intermediate state. The trade-off is that extra combinational logic is now required to work out, *before* each clock edge arrives, precisely which flip-flops should toggle on that edge and which should not.

Construction of a 4-bit synchronous binary up-counter using J-K flip-flops:

- All four flip-flops are **positive-edge-triggered J-K flip-flops**, and all four flip-flops' clock inputs are wired directly to the same common **Clock** line.
- A **Count Enable** signal is provided; while Count Enable is 1, the counter is active and free to count on each clock pulse (setting Count Enable to 0 freezes the counter, since it forces every flip-flop's J and K inputs to 0, i.e. "no change").
- The first flip-flop's J and K inputs are tied directly to **Count Enable**, so Q_0 toggles on *every* clock pulse whenever the counter is enabled (exactly as in the ripple counter).
- Each subsequent flip-flop's J and K inputs are driven by an **AND gate** that combines Count Enable with the outputs of *all* the less-significant flip-flops, so that a given flip-flop only toggles on the next clock edge if *every* bit below it is currently 1 (exactly the condition for a binary ripple carry to reach that bit position):
  - $J_1 = K_1 = \text{Count Enable} \cdot Q_0$
  - $J_2 = K_2 = \text{Count Enable} \cdot Q_0 \cdot Q_1$
  - $J_3 = K_3 = \text{Count Enable} \cdot Q_0 \cdot Q_1 \cdot Q_2$
- In practice this multi-input AND condition is not built as one large gate per stage; instead it is built **incrementally**, using a chain of small 2-input AND gates, where each AND gate's output feeds into the *next* AND gate together with the next Q output – e.g. the gate producing J_2/K_2 takes as its two inputs: (a) the output of the AND gate that produced J_1/K_1, and (b) Q_1. This is more hardware-efficient than a separate wide AND gate for every stage.

Circuit and timing diagram (4-bit synchronous binary counter, 9th clock pulse shown):



Step-by-step process on each clock pulse:

1. **Before** the active (positive) clock edge arrives, the AND-gate network has already settled and is continuously presenting the correct J, K values to every flip-flop, based on the *current*

values of $Q_0, Q_1, Q_2, \ldots$ and Count Enable.

2. When the positive clock edge arrives, it reaches *all four* flip-flops at the same instant (since they share one common clock line).
3. Each flip-flop reacts according to its own J, K inputs at that instant: if $J = K = 1$ (toggle condition satisfied) it flips; if $J = K = 0$ it holds its current value.
4. All of this happens **simultaneously** across every stage, so the entire counter value updates in a single, uniform step – unlike the ripple counter, where each stage only updates after seeing the previous stage's transition.

Worked trace of the first few pulses (Count Enable = 1 throughout):

- **Initially**: $Q_3Q_2Q_1Q_0 = 0000$. Only $J_0 = K_0 = 1$ (tied to Count Enable); all other J, K pairs are 0 because they depend on Q's that are currently 0.
- **1st clock pulse**: only Q_0 toggles ($0 \to 1$); Q_1, Q_2, Q_3 all have $J = K = 0$ so they hold. Result: 0001.
- **2nd clock pulse**: now $Q_0 = 1$, so $J_1 = K_1 = 1$ as well; both Q_0 and Q_1 toggle simultaneously on this same edge. Result: 0010.
- **3rd clock pulse**: Q_0 toggles again (always does); Q_1's toggle condition needs $Q_0 = 1$, which was true just before this edge, so Q_1 also toggles. Result: 0011.
- **4th clock pulse**: now $Q_0 = Q_1 = 1$ beforehand, so $J_2 = K_2 = 1$ as well; Q_0, Q_1 and Q_2 *all* toggle on this single edge simultaneously. Result: 0100.
- This pattern continues, matching the same binary count sequence as the ripple counter ($0000, 0001, 0010, \ldots, 1111$, then overflow back to 0000) – the *final count sequence produced is identical* to a ripple counter's; only the *internal timing* of how each bit updates differs.

Advantages of synchronous counters:

- **No cumulative propagation delay**: since every flip-flop is clocked at the same instant, the counter's total delay per clock pulse is just a single flip-flop's propagation delay (plus the settling time of the AND-gate network), *not* $n \times t_{pd}$ – this allows synchronous counters to operate at much higher maximum clock frequencies than ripple counters of the same bit-width.
- **No momentary invalid/glitch states**: because all bits update together, the counter's output is never briefly "wrong" partway through a transition, which makes synchronous counters safe to decode or act on immediately after every clock edge.
- Scales much better to counters with many bits, which is why synchronous counters are preferred in high-speed digital systems (e.g. inside a CPU).

Disadvantages of synchronous counters:

- Requires significantly more hardware than a ripple counter: an extra AND-gate network (growing with the number of bits) is needed to generate the correct J, K (or equivalent) inputs for every stage.
- More complex to design, since the enabling logic for every additional bit must be derived and wired correctly.
- Slightly higher power consumption than a ripple counter of the same size, due to the extra combinational logic switching on every clock cycle.

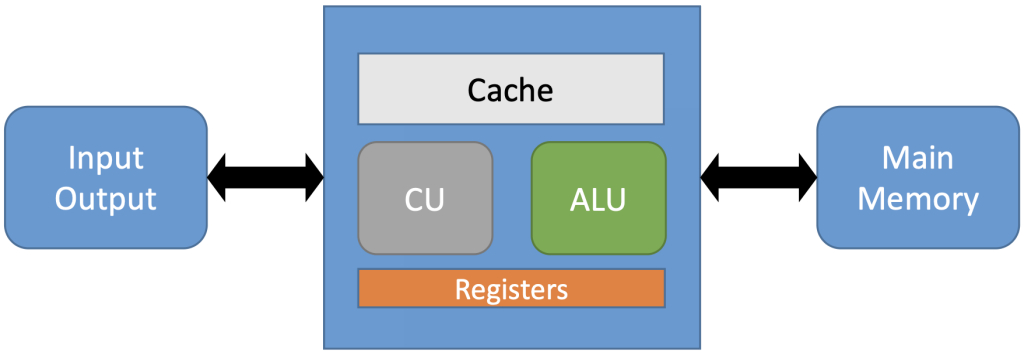
5 The CPU and Its Organization

5.1 Key Components

- **ALU (Arithmetic Logic Unit)**: performs mathematical and logical operations.
- **CU (Control Unit)**: fetches instructions from main memory, decodes them, determines which operations the ALU should perform, and controls the flow of data through the system.
- **Registers**: small, very fast storage locations that can hold binary information – much faster to access than main memory. They store a wide variety of data such as instructions, data values and addresses. Key registers include:

Register	Name	Purpose
MAR	Memory Address Register	Holds the address of the memory location to be accessed.
MDR	Memory Data Register	Holds data being transferred to or from memory.
AC	Accumulator	Stores intermediate arithmetic and logic results.
PC	Program Counter	Contains the address of the next instruction to execute.
CIR	Current Instruction Register	Contains the current instruction while it is being processed/decoded.

- **Cache**: a small, very fast memory that stores the most frequently (or recently) used data, to reduce the average time needed to access memory.
- **Input/Output interfaces**: allow the CPU to communicate with peripheral devices.



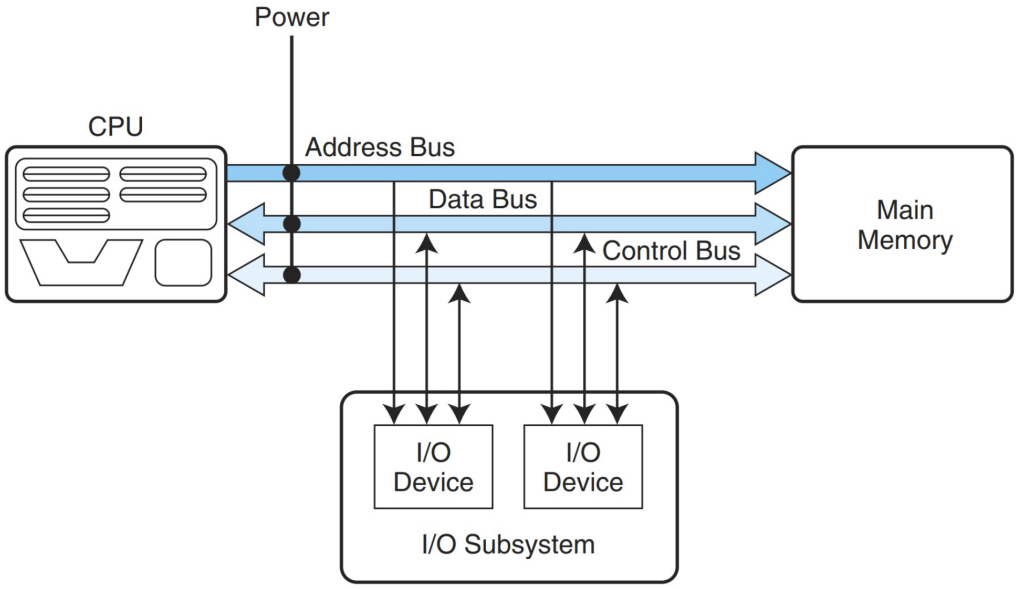
5.2 Main Memory and Addressing

- Main memory is organised as a sequence of individually addressable storage locations.
- **Byte-addressable** memory: each individual byte has its own unique address (the most common scheme in modern systems).
- **Word-addressable** memory: each address refers to one whole word (which may span several bytes), rather than each individual byte.
- The number of bits in an address determines the maximum number of distinct memory locations that can be addressed: with n address bits, up to 2^n locations can be addressed.

Worked example: A machine has a 256-byte main memory. Since $256 = 2^8$, an address needs at least 8 bits to uniquely identify every byte of memory.

5.3 Buses

- A **bus** is a shared set of wires that connects multiple subsystems, acting as a path along which data can travel between them.
- Only one device can use a given bus at any instant, since sharing it among multiple simultaneous transfers would create bottlenecks/collisions.
- Bus speed (effective bandwidth) depends on both the physical **width** of the bus (number of parallel wires/bits it can carry at once) and the number of devices that must share it.



- **Address Bus**: carries the address of the memory location (or device) that the CPU wants to read from or write to. It is **unidirectional** (one-directional): addresses only ever travel from the CPU outward.
- **Data Bus**: carries the actual data being transferred. It is **bidirectional**: if the CPU issues a read request, data flows from main memory to the CPU; if the CPU issues a write request, data flows from the CPU to main memory.
- **Control Bus**: carries control signals, such as read/write command lines, interrupt requests, and other coordination signals. It is bidirectional.

5.4 Clock and Clock Speed

- The CPU's **clock** generates a regular sequence of pulses that synchronise the operations of a synchronous digital system.
- **Clock speed** (clock rate) is measured in Hertz (Hz), i.e. cycles per second (commonly given in GHz, billions of cycles per second, for modern processors).
- **Clock cycle time** is the reciprocal of clock speed: $T = 1/f$. A higher clock speed means a shorter clock cycle time, and (generally) faster execution – though actual performance also depends heavily on how many clock cycles each instruction requires.

5.5 CPU Performance

- **CPU time** (execution time) for a program can be expressed as:

$$\text{CPU Time} = \text{Instruction Count} \times \text{CPI} \times \text{Clock Cycle Time}$$

or equivalently

$$\text{CPU Time} = \frac{\text{Instruction Count} \times \text{CPI}}{\text{Clock Rate}}$$

where **CPI** (Cycles Per Instruction) is the average number of clock cycles required to execute one instruction.

- **Instruction count** can be considered in two ways:
  - **Static instruction count**: the number of distinct instructions that appear in the compiled program (the size of the program's machine code).
  - **Dynamic instruction count**: the number of instructions actually *executed* while the program runs (which can differ substantially from the static count due to loops and branches being executed multiple times, or code paths being skipped).
- If a program's instruction mix consists of several instruction types, each with its own CPI, then the overall (average) CPI is the weighted average:

$$\text{CPI} = \sum_i \left(\text{CPI}_i \times \frac{\text{Instruction Count}_i}{\text{Total Instruction Count}} \right)$$

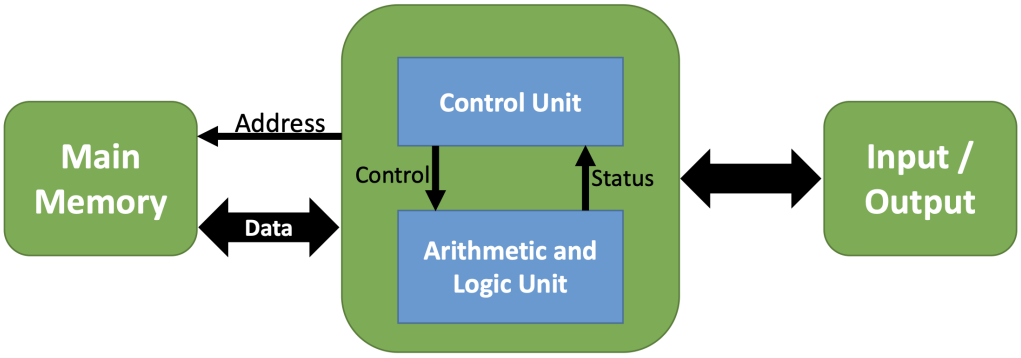
- *Worked example:* Suppose a program has three instruction types with the following counts and CPIs – Type A: 20,000 instructions at CPI 3; Type B: 30,000 instructions at CPI 4; Type C: 50,000 instructions at CPI 5 – and the clock rate is 2 GHz.

$$\begin{aligned}
\text{Total instruction count} &= 20{,}000 + 30{,}000 + 50{,}000 = 100{,}000 \\
\text{Total cycles} &= (20{,}000 \times 3) + (30{,}000 \times 4) + (50{,}000 \times 5) \\
&= 60{,}000 + 120{,}000 + 250{,}000 = 430{,}000 \text{ cycles} \\
\text{Average CPI} &= \frac{430{,}000}{100{,}000} = 4.3 \\
\text{CPU Time} &= \frac{430{,}000}{2 \times 10^9} = 215\ \mu\text{s}
\end{aligned}$$

- Reducing CPU time can be achieved by: reducing the instruction count (better compiler optimisation or a more efficient algorithm/ISA), reducing the average CPI (a more efficient microarchitecture, e.g. pipelining), or reducing the clock cycle time (increasing the clock rate, subject to physical/power limits).

5.6 Von Neumann Architecture

The **Von Neumann Architecture** refers to a stored-program computer design – proposed by the mathematician John von Neumann in 1945 – in which both instructions and data are stored together in the same main memory and are fetched and executed sequentially by the CPU via a single shared data path (often called the **von Neumann bottleneck**, since only one word can travel along it at a time). It organises a computer around three principal blocks – Main Memory, the CPU (containing the Control Unit and the Arithmetic and Logic Unit), and Input/Output – connected as shown below.



- The **Address** line carries the memory location the CPU wants to access (sent from the CPU to Main Memory).
- The **Data** line carries data between Main Memory and the CPU, and is bidirectional.
- Within the CPU, the **Control Unit** sends **Control** signals to the ALU telling it what operation to perform, and receives **Status** signals back from the ALU (e.g. flags indicating overflow, zero result, etc.).
- Input/Output devices connect to the rest of the system and exchange data with it.

5.7 The Fetch-Decode-Execute Cycle

Every instruction executed by the CPU passes through this repeating cycle:

1. **Fetch**: the CPU fetches the next instruction to execute from memory (using the address held in the Program Counter), and loads it into the appropriate register (the Current Instruction Register).
2. **Decode**: the CPU decodes the fetched instruction; any required operands are fetched from memory and placed into registers.
3. **Execute**: the CPU executes the instruction (e.g. the ALU performs the required operation).
4. The CPU **updates the Program Counter** (normally incrementing it to point at the next instruction, unless the executed instruction was itself a jump/branch).
5. The cycle **repeats** from the top for the next instruction.

5.8 Instruction Set Architecture (ISA)

The **Instruction Set Architecture** is the set of machine-specific instructions that a given computer/processor can execute, together with the precise format (encoding) of each instruction.

5.8.1 Types of Machine Instructions

Machine instructions generally fall into three broad categories:

- **Data Transfer instructions**: transfer data between registers and memory cells.
- **Arithmetic/Logic instructions**: perform operations such as addition, AND, OR, XOR, etc.
- **Control instructions**: control the execution flow of the program (e.g. jumps, halting).

Data Transfer Instructions – example mnemonics:

Instruction	Meaning
`L R, A`	LOAD register R with the content of memory cell A.
`LI R, I`	LOAD register R with the immediate value I (a literal number given directly in the instruction).
`ST R, A`	STORE the content of register R to the memory cell whose address is A.
`LR R1, R2`	LOAD register $R1$ with the content of register $R2$.

Worked example – swap the contents of two memory cells `0x30` and `0x40`:

```
L R1, 30    Load R1 with the content in memory cell 30
L R2, 40    Load R2 with the content in memory cell 40
ST R1, 40   Store R1 (originally cell 30's value) to cell 40
ST R2, 30   Store R2 (originally cell 40's value) to cell 30
```

Arithmetic/Logic Instructions – example mnemonics:

Instruction	Meaning
`ADD R0, R1, R2`	ADD the numbers in $R1$ and $R2$ (represented in two's complement) and place the result in $R0$.
`AFP R0, R1, R2`	ADD the numbers in $R1$ and $R2$ (represented in floating-point) and place the result in $R0$.
`OR R0, R1, R2`	OR the bit patterns in $R1$ and $R2$; result in $R0$.
`AND R0, R1, R2`	AND the bit patterns in $R1$ and $R2$; result in $R0$.
`XOR R0, R1, R2`	XOR the bit patterns in $R1$ and $R2$; result in $R0$.

Worked example – mask the first 4 bits of the binary string stored at memory cell `A0` (using `AND` with a mask of `0F` $= 00001111_2$ to clear the upper nibble and keep only the lower 4 bits):

```
L R1, 80        Load R1 with the content of cell 80 (the data to mask)
LI R2, 0F       Load R2 with the immediate mask value 0F
AND R0, R1, R2  AND R1 and R2, giving only the lower 4 bits in R0
ST R0, A0       Store the masked result R0 back to cell A0
```

Control Instructions – example mnemonics:

Instruction	Meaning
`JMP R, A`	JUMP to the instruction at memory cell A *if* the bit pattern in R equals the one in $R0$ (if equal, reset the Program Counter to address A).
`HALT`	Halt execution.

Worked example – a small control-flow loop (using registers to accumulate a running addition, with a conditional jump forming the loop, and `LR` copying a register for the next comparison):

```
LI R0, 0A        JMP R3, 3E
LI R1, 00        LR R1, R3
LI R2, 01        JMP R0, 36
ADD R3, R1, R2   HALT
```

This repeatedly adds $R2$ to the running total in $R1$ (via $R3$), jumping back to re-execute the `ADD` until the accumulated value in $R3$ equals $R0$ (`0A`), at which point the conditional jump to `3E` succeeds and the program halts – illustrating how conditional jumps and register copying together implement a loop at the machine-instruction level.

5.8.2 Case Study: Designing a 16-bit Instruction Format

Consider a specific hypothetical architecture used to illustrate how an ISA is actually designed within a fixed instruction width:

- Word-addressable memory (word size = 16 bits).
- Instructions are 16 bits long, split into: 4 bits for the **Opcode**, and 4 bits used for a **register identifier** (since $2^4 = 16$ registers can be uniquely addressed with 4 bits).
- Addresses are 8 bits long.

Since a single instruction is only 16 bits wide in total, and different instructions need different combinations of operands (an immediate value, a memory address, one register, two registers, or three registers), **a single fixed layout cannot efficiently serve every instruction** – so the ISA defines several different **instruction layouts (types)**, each still exactly 16 bits wide in total, sharing a common 4-bit Opcode field at the start but differing in how the remaining 12 bits are divided.

5.8.3 Instruction Layout: Type 1 (Register + Immediate Value)

$$\underbrace{\text{Opcode (4 bits)}}\ \underbrace{\text{Register (4 bits)}}\ \underbrace{\text{Immediate Value (8 bits)}}$$

Opcode	Instruction	Meaning
`0x2`	`LI R, I`	Load Immediate
`0xA`	`RL R, I`	Rotate Left
`0xB`	`RR R, I`	Rotate Right
`0xC`	`SL R, I`	Shift Left
`0xD`	`SR R, I`	Shift Right

5.8.4 Instruction Layout: Type 2 (Register + Memory Address)

$$\underbrace{\text{Opcode (4 bits)}}\ \underbrace{\text{Register (4 bits)}}\ \underbrace{\text{Memory Address (8 bits)}}$$

Opcode	Instruction	Meaning
`0x1`	`L R1, 0x11`	Load from a memory address
`0x3`	`ST R2, 0x12`	Store at a memory address
`0xE`	`JMP R3, 0x11`	Conditional jump

Note that Type 2 has *exactly the same bit widths* as Type 1 (Opcode/Register/8-bit field); the only difference is how that final 8-bit field is *interpreted* – as a memory address rather than as an immediate literal value. This is a common ISA design pattern: reusing an identical bit layout for multiple instruction types and letting the Opcode alone determine the correct interpretation of the remaining fields.

5.8.5 Instruction Layout: Type 3 (Three Registers)

$$\underbrace{\text{Opcode (4 bits)}}\ \underbrace{\text{Register (4 bits)}}\ \underbrace{\text{Register (4 bits)}}\ \underbrace{\text{Register (4 bits)}}$$

Opcode	Instruction	Meaning
0x5	ADD R1,R2,R3	Addition ($R1 = R2 + R3$)
0x6	AFP R1,R2,R3	Floating point addition
0x7	OR R1,R2,R3	Logical OR
0x8	AND R1,R2,R3	Logical AND
0x9	XOR R1,R2,R3	Logical XOR

Here the 12 bits remaining after the Opcode are split into three separate 4-bit register fields (a destination register and two source registers), since all these operations need three register operands and no memory address or immediate value.

5.8.6 Instruction Layout: Type 4 (Two Registers, With Unused Bits)

$$\underbrace{\text{Opcode (4 bits)}}\quad\underbrace{\text{Unused (4 bits)}}\quad\underbrace{\text{Register (4 bits)}}\quad\underbrace{\text{Register (4 bits)}}$$

Opcode	Instruction	Meaning
0x4	LR R1, R2	Load Register ($R1 = R2$)

This instruction only needs two register operands, but the fixed 16-bit format still allocates 12 bits after the Opcode; the unused 4 bits are simply padded with zeros (0000) rather than being reclaimed, since the instruction width must stay fixed at 16 bits for every instruction regardless of type.

5.8.7 Full Instruction Set (Example)

Combining Types 1–4 together with the earlier Control instruction gives the full opcode table for this machine (opcodes 0x0 and 0xF left free for additional/custom instructions, e.g. a hypothetical MULT R0,R1,R2):

Opcode	Instruction	Opcode	Instruction
1	L R, A	9	XOR R0,R1,R2
2	LI R, I	A	RL R, I
3	ST R, A	B	RR R, I
4	LR R1, R2	C	SL R, I
5	ADD R0,R1,R2	D	SR R, I
6	AFP R0,R1,R2	E	JMP R, A
7	OR R0,R1,R2	F	HALT
8	AND R0,R1,R2	0	(additional instruction, e.g. MULT R0,R1,R2)

5.8.8 Worked Example: Assembler to Machine Code

Encoding the earlier "swap two memory cells" program using this instruction set (Opcode in the high nibble, Register in the next nibble, Address in the low byte – Type 2 layout):

Assembler	Machine Code (binary)	Hex
L R1, 30	0001 0001 0011 0000	1130
L R2, 40	0001 0010 0100 0000	1240
ST R1, 40	0011 0001 0100 0000	3140
ST R2, 30	0011 0010 0011 0000	3230

Reading `1130` as an example: the first hex digit `1` is the Opcode (`L`, load), the second hex digit `1` is the Register field (`R1`), and the remaining byte `30` is the 8-bit memory address – exactly matching the Type 2 layout (Opcode | Register | Memory Address).

6 Addressing Modes and Instruction-Level Pipelining

6.1 Addressing Modes

Addressing modes specify *where* the operand(s) of an instruction are located – an addressing mode can specify a constant, a register, or a location in memory. A typical instruction format reserves a fixed number of bits for the opcode, and the remaining bits are interpreted differently depending on the addressing mode – e.g. an opcode field followed by a register field and then either an immediate value, a memory address, or additional register fields. Real computer architectures typically provide *multiple* of the addressing modes below, so that the most efficient one can be chosen for each situation.

6.1.1 Effective Address and Physical Address

- The **effective address** of an operand is the location where the actual operand value can be found.
  - In a system *without* virtual memory, the effective address is either a main memory address or a register.
  - In a system *with* virtual memory, the effective address is a virtual address or a register.
- The **physical address** is the real hardware address in main memory. Translating from an effective/virtual address to the physical address is the responsibility of the **Memory Management Unit (MMU)**, a hardware device that manages and controls access to memory. Because the MMU handles this translation automatically, the process is invisible to the programmer.

6.1.2 Immediate Addressing

- The instruction itself contains the operand's actual value; the opcode is immediately followed by the value to be used, so the data item is available as part of the instruction.
- *Example*: `LI R1, 0x0A` – load register R1 with the immediate value `0x0A`.
- **Pros**: very fast, since the value is already included in the instruction (no extra memory access needed to fetch it).
- **Cons**: less flexible, since the value is fixed at compile time; the size of the representable number is restricted by the size of the instruction's address/operand field.

6.1.3 Direct Addressing

- The instruction's address field directly contains the effective address of the operand.
- *Example*: `LOAD R1, 0x80` – load register R1 with the contents of memory location `0x80`.
- **Pros**: fast, since the value can be accessed with a single memory reference; more flexible than immediate addressing since the value itself is not baked into the instruction.
- **Cons**: provides only a limited address space, since the address field width restricts how much memory can be directly addressed.

6.1.4 Indirect Addressing

- The instruction's address field contains a **pointer** to the effective address of the operand (i.e. the address field holds the address of the address).
- *Example*: given an array starting at address `0x1000`, with register R1 holding the base address (`0x1000`) and R2 holding an index (e.g. 2), the effective address of the desired element is first computed: `ADD R3, R1, R2`. The value at that computed address is then loaded using indirect addressing: `LD R4, (R3)` – the parentheses around `R3` indicate a reference to the value pointed to by the address stored in R3 (directly analogous to pointer dereferencing in C, e.g. `int value = *ptr;`).
- **Pros**: allows access to a larger memory space, since an entire word/register can hold a full address.
- **Cons**: executing the instruction requires **two** memory references – one to fetch the address, and a second to fetch the actual operand value – making it slower than direct addressing.

6.1.5 Register Addressing (Register Direct)

- Similar in concept to direct addressing, but a **register** (rather than a memory location) holds the operand.
- *Example*: `ADD R1, R2, R3` – all operands are registers.
- **Pros**: very fast, since accessing a register is much quicker than accessing memory; no memory reference is required at all.
- **Cons**: limited by the small number of available registers, restricting how much data can be held this way at once.

6.1.6 Register Indirect Addressing

- A variant of indirect addressing: the instruction's operand specifies a **register** that holds a memory address (the effective address) of the actual data item to be used (rather than the address field of the instruction pointing to a memory location holding a pointer).
- **Pros**: only a single memory reference is required to fetch the operand (faster than full indirect addressing, since the pointer itself is already in a register rather than needing to be fetched from memory).
- **Cons**: still limited address space compared to full indirect addressing, since it relies on a fixed number of registers to hold pointers.

6.1.7 Displacement Addressing

- A combination of direct addressing and register indirect addressing.
- The instruction contains **two** address fields: one explicit (a direct address or constant offset) and one implicit/register-based; the effective address is calculated by adding the register's contents to the given displacement/offset value.
- This is the basis of common addressing techniques such as *indexed addressing* and *base-register addressing* used to access arrays and structures efficiently.

6.1.8 Stack Addressing

- The operand is implicitly located on top of a hardware **stack**, accessed via a dedicated Stack Pointer register, rather than through an explicit address field in the instruction.
- Common in architectures that support subroutine calls (storing return addresses) and expression evaluation.

6.1.9 Worked Comparison Example

Suppose memory location `0x100` contains the value `0x210`; location `0x210` contains the value `0x350`; and location `0x350` contains the value `0x050`. For the instruction `LOAD R1, 0x100`:

Addressing Mode	**Value loaded into R1**	**Explanation**
Immediate	`0x100`	The literal value given in the instruction itself.
Direct	`0x210`	The contents of memory location `0x100`.
Indirect	`0x350`	The contents of the location *pointed to by* `0x100` – i.e. the contents of `0x210`.

6.2 Instruction-Level Pipelining

- Modern CPUs divide the fetch–decode–execute cycle into a series of small, independent steps.
- Because these steps are made independent of one another, different steps *belonging to different instructions* can be carried out in parallel – this technique is called **pipelining**.
- The core idea: divide the execution of a single instruction into several sequential stages, then execute the corresponding stage of several *different* (consecutive) instructions simultaneously, much like an assembly line.

An example set of pipeline stages:

1. Fetch instruction
2. Decode opcode
3. Calculate the effective address of the operand(s)
4. Fetch operand(s)
5. Execute instruction
6. Store result

Instruction-level pipelining overlaps these stages across a set of consecutive instructions (e.g. while instruction 2 is being decoded, instruction 1 can already be having its effective address calculated, and instruction 3 can simultaneously begin being fetched). For maximum throughput, the time taken by each pipeline stage should ideally be balanced (roughly equal), since the overall pipeline can only advance as fast as its slowest stage.

6.2.1 Pipelining Efficiency and Speedup

Let there be a k**-stage** pipeline, with clock cycle time t_p (the time taken per stage), and n instructions to be processed.

- The first instruction, I_1, must pass through all k stages before it completes, requiring $k \times t_p$ time.

- Because the pipeline is now full, each of the remaining $n-1$ instructions emerges (completes) from the pipeline at a rate of one per clock cycle, taking a further $(n-1)t_p$ time in total.
- **Total time to complete n instructions using a k-stage pipeline:**

$$k \times t_p + (n-1)t_p = (k+n-1)\,t_p$$

Speedup compares the pipelined execution time to the time a non-pipelined CPU would take (which would need $k \times t_p$ time *for every* instruction, i.e. $k \times t_p \times n$ total):

$$\text{Speedup} = \frac{k \times t_p \times n}{(k+n-1)\,t_p} = \frac{k \cdot n}{k+n-1}$$

As n becomes very large, this speedup approaches k (the number of pipeline stages) – i.e. the theoretical maximum benefit of pipelining is proportional to the number of stages, though this ideal is rarely fully achieved in practice.

6.2.2 Advantages, Disadvantages, Limitations and Issues

- **Advantages**: significantly increases instruction throughput (instructions completed per unit time) without requiring a faster clock; makes efficient use of CPU hardware resources, since different functional units can be kept busy processing different instructions' stages simultaneously.
- **Disadvantages**: significantly increases hardware design complexity; latency for any single instruction is not reduced (an individual instruction still takes k stages to complete) – pipelining improves throughput, not the completion time of one instruction.
- **Limitations**: the pipeline can only run as fast as its *slowest* stage, so stage durations must be carefully balanced; deeper pipelines (larger k) give diminishing practical returns due to overheads.
- **Issues (hazards)**: pipelining introduces **hazards** that can stall or disrupt the pipeline, such as:
  - **Structural hazards**: two instructions in different stages need the same hardware resource at the same time.
  - **Data hazards**: an instruction depends on the result of a previous instruction that has not yet completed/written back its result.
  - **Control hazards**: caused by branch/jump instructions, where the next instruction to fetch is not known until the branch itself has been resolved, potentially wasting the work already done on incorrectly fetched instructions.

7 Memory Organization

7.1 Primary and Secondary Memory

- **Primary Memory**: memory directly accessible to the CPU.
  - Cache Memory
  - Main Memory
- **Secondary Memory**: memory used for longer-term, larger-capacity storage, not directly accessed by the CPU.

  - Magnetic Disks (Hard Disk Drives)
  - Magnetic Tapes
  - USB Sticks / Flash Drives

7.2 Types of Memory

- **Read Only Memory (ROM)**: **non-volatile** (retains its content without power). Variants include:
  - **PROM** (Programmable Read Only Memory)
  - **EEPROM** (Electrically Erasable Programmable Read Only Memory)
- **Random Access Memory (RAM)**: **volatile** (loses its content when power is removed). Variants include:
  - **DRAM** (Dynamic RAM)
  - **SRAM** (Static RAM)

7.3 Flash Memory

- Built from floating-gate metal-oxide-semiconductor field-effect transistors.
- **Non-volatile**, and supports both reading and writing.
- Common uses: memory cards, USB flash drives, solid state drives.

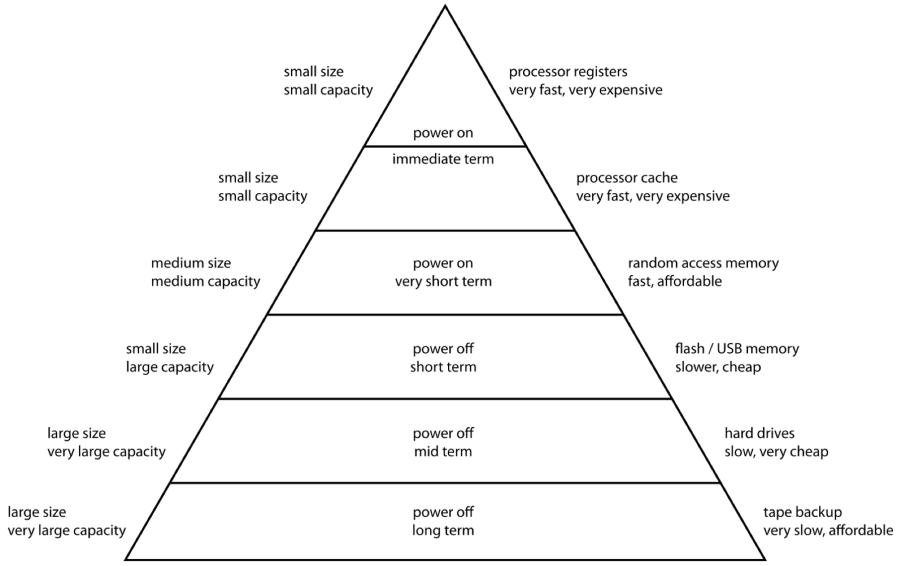
7.4 Solid State Devices (SSD)

- Typically built on flash memory technology.
- Compared to traditional Hard Disk Drives (HDD): more resistant to physical shock, run silently (no moving parts), and offer faster access times.
- **Not** well suited for very long-term storage without power, since charge stored in the memory cells depletes gradually over time.

(3D XPoint is mentioned as a newer memory technology worth further independent reading, positioned between DRAM and flash in terms of speed/persistence characteristics.)

7.5 Memory Hierarchy

Computer storage components are organised into a **hierarchy** based on several attributes: **access time / latency**, **cost** (per unit of storage), **size** (capacity), and **distance from the CPU**. As one moves up the hierarchy (towards the CPU), memory becomes faster and more expensive per byte, but smaller in capacity; moving down, memory becomes slower and cheaper per byte, but much larger in capacity.



From top to bottom of the pyramid: processor registers (smallest, fastest, most expensive, immediate/volatile on power-off) → processor cache → main memory (RAM) → flash/USB memory → hard drives → tape backup (largest, slowest, cheapest, persists long-term without power).

7.6 Typical Data Access Pattern

1. The CPU first requests data from its nearest storage: the **cache**.
2. If the data is not found in the cache, the request proceeds to **main memory**.
3. If still not found, the request proceeds to secondary storage (e.g. **HDD**).
4. Once the data is located, it is loaded (copied) into main memory, and then into cache, so that it (and often, by locality, nearby data) is readily available for subsequent accesses.

7.7 The CPU–Memory Gap

- CPU cycle time has historically decreased (i.e. CPU speed increased) much more rapidly than memory access times have improved.
- Typical relative speed ordering: HDD > SSD > DRAM > SRAM $\gg$ CPU (in terms of access latency – HDD is by far the slowest, the CPU is by far the fastest).
- This growing mismatch between CPU speed and memory speed is called the **CPU–Memory Gap**, and it can significantly limit overall system performance if the CPU is frequently left waiting for data.
- The **Principle of Locality** is exploited to help bridge this gap.

7.8 Principle of Locality

The principle of locality describes the tendency of a CPU to access the **same set of memory locations repeatedly over a short period of time**: programs tend to reference data and instructions with addresses that are near, or equal, to addresses they have referenced recently.

- **Spatial Locality**: items with nearby addresses tend to be referenced close together in time (e.g. iterating sequentially through an array).

- **Temporal Locality**: a recently referenced item is likely to be referenced again in the near future (e.g. a loop variable or a frequently-called function).

Caching exploits both forms of locality: bringing in a whole block of nearby data exploits spatial locality, while keeping recently used data in the (small, fast) cache exploits temporal locality.

7.9 Cache Memory

- A cache is a smaller, faster memory space that holds (a copy of) data from a larger, slower memory space.
- Typically resides physically on the CPU chip itself.
- Typical characteristics: **transient** (highly volatile – its content is not meant to persist); may hold frequently referenced data, or data predicted to be referenced in the near future.

7.9.1 Cache Hierarchy

- Cache memories are typically built from **SRAM**, which has a high per-unit cost.
- Because larger and faster cache space is expensive, a **hierarchy of caches** is used instead of one large cache – different levels of cache (e.g. **L1**, L2, L3), where L1 is the smallest and fastest (closest to the CPU core), and each successive level is larger but slower.

7.10 Cache Performance

- **Cache Hit**: the data requested by the CPU is found in the cache.
- **Cache Miss**: the requested data is *not* found in the cache.
- **Cache Hit Ratio (HR)**: the probability that a requested item is found in the cache.
- **Cache Miss Ratio**: 1 − Hit Ratio.
- **Effective Access Time (EAT)**: the average time it takes for the CPU to access data, taking into account both hits and misses:

$$EAT = HR \times T_h + (1 - HR) \times T_m$$

  where HR is the Hit Ratio, T_h is the cache access time on a hit, and T_m is the (typically much larger) data access time on a miss (which usually also includes the time to then access main memory).

7.11 Key Design Decisions of a Cache

1. **Cache Volume** (capacity / size).
2. **Write Policy**: Write Back or Write Through.
3. **Mapping Scheme**: Direct Mapped, Fully Associative, or Set Associative (including choice of line/block replacement policy where relevant).

7.11.1 Write Policy

Because a cache is a form of **redundant storage** – the same data conceptually exists in both the cache and in main memory (and may also be shared across multiple processes) – maintaining **consistency** between the cache and main memory becomes a challenge whenever the data is altered (written to). A cache therefore needs a clearly defined write policy:

- **Write Through**: whenever a write operation happens, it is performed on *both* the cache and main memory simultaneously (synchronously).
  - Comparatively simple to implement.
  - Increases bus traffic, since every single write generates a main-memory transaction.
- **Write Back**: a write operation is performed *only* on the cache; the updated data is written back to main memory only later, when that cache entry is eventually **evicted** from the cache.
  - Sometimes called "lazy writes".
  - More complex to implement (the cache must track which entries have been modified, typically via a "dirty bit"), but reduces bus traffic since repeated writes to the same location only cost one main-memory write (on eviction), not one per write.

7.11.2 Mapping Schemes – Why They Are Needed

The CPU refers to data/instructions by their addresses. Data is searched for in the cache first, so cached data must be retrievable via addresses. However, since the cache is much smaller than main memory, cache locations cannot simply be given the same addresses as main memory locations – a **mapping scheme** is required to translate between a main-memory address and a location within the (much smaller) cache. The three main mapping schemes are:

- Direct Mapped
- Fully Associative
- Set Associative

7.11.3 Cache Lines (Cache Blocks) and Cache Entries

- Data is transferred between main memory and cache in fixed-size blocks called **cache lines** (or cache blocks) – not one byte/word at a time – which exploits spatial locality (nearby data is brought in together).
- When a cache line is copied into the cache, a **cache entry** is created, consisting of the copied data *plus* a **tag** that identifies which memory location(s) this entry corresponds to.
- The amount of data held in a cache entry is always exactly equal in size to a cache line.

7.12 Direct Mapped Cache

- Main memory is conceptually divided into fixed-size blocks.
- Each memory block has a single, pre-defined cache-entry position that it can be placed into; i.e. a given memory block will *always* map to the same cache entry position.
- Since the number of blocks in main memory is far larger than the number of available cache entries, several different memory blocks necessarily map to the *same* cache entry position, and only one of them can occupy that position at any given time.
- A **tag** is stored alongside each cache entry to identify exactly which memory block currently occupies that position.

7.12.1 Address Breakdown: Tag, Index, Offset

A memory address used to access a direct-mapped cache is conceptually split into three fields:

- **Offset**: the low-order bits identifying which specific byte/word *within* a cache line is being referred to.
- **Index**: the next bits, identifying *which* cache entry position (line slot) the address maps to.
- **Tag**: the remaining (high-order) bits, stored with the cache entry, used to verify *which* of the several possible memory blocks currently occupies that cache entry.

Worked example (small illustrative cache): A byte-addressable main memory of 1024 bytes, using a direct-mapped cache with 2 bytes per cache line and a total cache volume of 8 bytes (so there are $8/2 = 4$ cache entry positions).

- Since each cache line holds 2 bytes, **1 bit** is needed for the Offset (to select between the 2 bytes within a line).
- Since there are 4 cache entry positions, **2 bits** are needed for the Index (to select 1 of 4 positions).
- Since main memory is 1024 bytes, a full address needs $\log_2(1024) = 10$ bits; subtracting the 1 offset bit and 2 index bits leaves $10 - 1 - 2 = 7$ bits for the Tag.
- So an address is split as: `Tag (7 bits)` | `Index (2 bits)` | `Offset (1 bit)`.

Searching a direct-mapped cache:

1. Use the **Index** bits of the requested address to locate the corresponding cache entry position directly.
2. Use a comparator to check whether the **Tag** stored at that cache entry matches the Tag bits of the requested address.
3. If the tag matches, check the entry's **Valid Bit**:
   - Valid bit = 1: **Cache Hit**.
   - Valid bit = 0: **Cache Miss** (the entry has never been validly loaded).
4. If the tag does *not* match, it is a **Cache Miss** (a different memory block currently occupies that position, so the requested block must be fetched from main memory and will replace/evict the current occupant of that position).

7.12.2 General Formulas for Tag / Index / Offset Lengths

$$\text{Offset length (bits)} = \log_2\left(\frac{\text{Cache line size}}{\text{Size of one addressable unit}}\right)$$

$$\text{Index length (bits)} = \log_2\left(\frac{\text{Total cache volume}}{\text{Cache line size}}\right)$$

$$\text{Tag length (bits)} = \text{Total address length (bits)} - \text{Index length} - \text{Offset length}$$

Worked exam-style example: a 2 MB direct-mapped cache, with a cache line size of 32 KB, for a **word-addressable** 1 GB main memory where each word is 32 bits (4 bytes).

- **Offset**: the number of addressable words (not bytes, since memory is word-addressable) within one cache line:

$$\frac{32\text{KB}}{32 \text{ bits}} = \frac{32 \times 2^{10} \text{ Bytes}}{4 \text{ Bytes}} = \frac{2^{15} \text{ Bytes}}{2^2 \text{ Bytes}} = 2^{13} \Rightarrow \textbf{13 bits for Offset}.$$

- **Index**: the number of cache entries (line slots) available:

$$\frac{\text{2MB}}{\text{32KB}} = \frac{2 \times 2^{20} \text{ Bytes}}{2^5 \times 2^{10} \text{ Bytes}} = \frac{2^{21}}{2^{15}} = 2^6 \Rightarrow \textbf{6 bits for Index}.$$

- **Total address length**: number of distinct word-addresses in a 1 GB, word-addressable (4 bytes/word) memory:

$$\frac{\text{1GB}}{\text{32 bits}} = \frac{2^{30} \text{ Bytes}}{\text{4 Bytes}} = 2^{28} \Rightarrow \textbf{28 bits total address length}.$$

- **Tag length** $= 28 - 6 - 13 =$ **9 bits**.

28 bits (total address)		
Tag Length	Index Length	Offset Length
9	6	13

Note: if the memory in such a question is **byte-addressable** instead of word-addressable, the offset/index/tag calculations are performed directly in bytes (no division by the word size is needed when computing the offset), which will generally shift how many bits are required for each field – always check carefully whether a question specifies byte or word addressability.

7.12.3 Advantages and Disadvantages of Direct Mapped Cache

- **Advantages**: simple to implement in hardware; fast lookups, since the index bits give the exact position to check with a single comparison (no searching required).
- **Disadvantages**: prone to a high number of cache misses if the program happens to repeatedly access two or more memory blocks that map to the *same* cache position (an effect sometimes called "thrashing"), even if the rest of the cache is empty/unused; is the least flexible mapping scheme.

7.13 Fully Associative Cache

- The direct-mapped scheme enforces a strict mapping rule (each block maps to exactly one position); **fully associative** caching is the complete opposite.
- There are no restrictions: an incoming cache line can be placed *anywhere* currently available in the cache.
- This is a much more flexible mapping scheme (far fewer unnecessary evictions/misses caused purely by mapping collisions).
- The major drawback is that **searching** the cache is expensive: since a block could be anywhere, every single cache entry's tag potentially needs to be checked (compared) on every access, requiring many parallel comparators.

7.13.1 Address Breakdown for a Fully Associative Cache

Because a cache line can be placed in *any* entry, there is no "Index" field at all – an address is split into only two fields:

- **Offset**: identifies which byte/word *within* a cache line is being referred to (exactly as before).
- **Tag**: *all* of the remaining (higher-order) address bits – i.e. every bit of the address other than the offset becomes part of the tag, since there is no index to narrow down the search first.

$$\text{Tag length (bits)} = \text{Total address length (bits)} - \text{Offset length (bits)}$$

Because the tag is comparatively much longer than in a direct-mapped or set-associative cache (there is no index to "absorb" some of the address bits), each individual cache entry requires more storage overhead purely for its tag.

7.13.2 Matching (Searching) Process in a Fully Associative Cache

1. The requested address is split into its **Tag** and **Offset** fields (no index bits exist to narrow the search).
2. The Tag field of the requested address is compared, **simultaneously and in parallel**, against the tag stored in *every single* cache entry, using a separate hardware **comparator** for each entry.
3. If *any* comparator reports a match *and* that entry's **Valid Bit** is set to 1, it is a **Cache Hit**: the matching entry's data, combined with the Offset field, gives the requested byte/word.
4. If *no* comparator reports a match (or the matching entry's Valid Bit is 0), it is a **Cache Miss**; the required block must be fetched from main memory and placed into any available (or selected via the replacement policy) cache entry.

Because every entry's tag must be compared at once, a fully associative cache of m entries needs m parallel comparators, each as wide as the full tag field – this hardware cost (in silicon area, wiring complexity, and power consumption) is the fundamental reason fully associative caches are only practical for relatively *small* caches (e.g. a small Translation Lookaside Buffer), rather than large, high-capacity caches.

7.13.3 Worked Example

A byte-addressable fully associative cache with total volume 8 bytes and 2 bytes per cache line, for a 1024-byte main memory (same base scenario as the direct-mapped example above, for direct comparison).

- **Offset**: 2 bytes per line $\Rightarrow$ **1 bit**.
- **Index**: none (0 bits) – there is no fixed position for any block.
- **Tag**: total address length is $\log_2(1024) = 10$ bits, so Tag $= 10 - 0 - 1 =$ **9 bits** (compared to only 7 bits of tag in the direct-mapped version of the same scenario, since here the tag must also absorb what was previously the 2-bit index).
- Number of cache entries $= 8/2 = 4$, so **4 comparators** (each 9 bits wide) are needed to search the whole cache in parallel.

7.13.4 Advantages and Disadvantages of Fully Associative Cache

- **Advantages**: maximum flexibility – a block can go anywhere, so conflict misses caused purely by two blocks mapping to the same fixed position (as can happen in a direct-mapped cache) are completely eliminated; makes the most efficient use of every available cache entry.
- **Disadvantages**: requires the most comparator hardware of any mapping scheme (one comparator per cache entry, all active on every single access), making it expensive, power-hungry, and slow to scale to large cache sizes; choosing which entry to evict when the cache is full also becomes more complex, since *any* entry is a candidate (the replacement policy must consider the entire cache, not just a small set).

7.14 Set Associative Cache

- A **hybrid** model, combining ideas from both fully associative and direct mapped caches.
- The cache is divided into a number of equal-sized **sets**.
- A given cache line maps to only *one* specific set (this part is **directly mapped**, just like direct-mapped caching, but mapping to a *set* rather than to one single position).
- Within its mapped set, however, the cache line can be placed *anywhere* available within that set (this part is **fully associative**, but only searching within the small set rather than the whole cache).
- If a set can accommodate N cache lines simultaneously, the scheme is called an N**-way set associative cache**. (A direct mapped cache is the special case $N = 1$; a fully associative cache is the special case where there is only 1 set containing every cache line, i.e. $N =$ total number of cache entries.)

Worked example: 2-way set associative cache. A byte-addressable, 2-way set associative cache, for a 1024-byte main memory, with 2 bytes per cache line and total cache volume of 8 bytes.

- Since it is 2-way, each set holds 2 cache lines, so the **set size** $= 2 \times 2$ bytes $= 4$ bytes.
- Since the total cache volume is 8 bytes, there are $8/4 = 2$ sets available.
- Address breakdown: an **Offset** field (to pick the byte within a line), a **Set Index** field (to pick which of the 2 sets a block maps to), and a **Tag** field (to identify which specific block, among those that map to the same set, currently occupies a given cache entry).
- Layout of a cache entry: `Tag | Set Index | Offset | Data | Valid Bit`.

Illustrative walk-through (same style as would appear in an exam): suppose the CPU successively references addresses 1, then 8, then 4 (or 5).

- Address 1 maps to Set 0; since the cache starts empty, it is placed in any free position within Set 0.
- Address 8 also maps to Set 0 (a different tag); since Set 0 can still accommodate a second cache line (2-way), it is placed alongside the entry for address 1 – *no eviction is needed*, unlike what would happen in a direct-mapped cache with only one slot per position.
- If the CPU then references address 4 or 5 (which also maps to Set 0), Set 0 is now full, so **one** of the two existing entries in Set 0 must be evicted to make room – *which* one is evicted is decided by the cache's **replacement policy**.

This illustrates the key benefit of set associativity over direct mapping: by allowing a small amount of flexibility (multiple ways per set), far fewer unnecessary evictions occur for blocks that happen to share the same index, at a smaller search cost than a fully associative cache (only the lines within one set need to be compared, not the whole cache).

7.14.1 General Formulas for Set-Associative Tag / Set Index / Offset Lengths

$$\text{Offset length (bits)} = \log_2\left(\frac{\text{Cache line size}}{\text{Size of one addressable unit}}\right)$$

$$\text{Number of sets} = \frac{\text{Total cache volume}}{\text{Set size}} = \frac{\text{Total cache volume}}{N \times \text{Cache line size}} \qquad \Rightarrow \qquad \text{Set Index length (bits)} = \log_2(\text{Numb}$$

$$\text{Tag length (bits)} = \text{Total address length (bits)} - \text{Set Index length} - \text{Offset length}$$

where N is the number of ways (cache lines per set). Notice that increasing N (more ways per set, for a fixed total cache volume) *decreases* the number of sets, which in turn *decreases* the Set Index length and correspondingly *increases* the Tag length – this is the general trade-off as a cache moves from direct-mapped ($N = 1$, largest index, smallest tag) towards fully associative (N = all entries, no index at all, largest tag).

7.14.2 Matching (Searching) Process in a Set-Associative Cache

1. Split the requested address into **Tag**, **Set Index**, and **Offset** fields.
2. Use the **Set Index** bits to directly locate exactly *one* set (exactly as in direct mapping, but identifying a whole set rather than a single entry).
3. Compare the requested **Tag** *simultaneously and in parallel* against the tags of *only the N entries within that one located set* (not the whole cache), using N parallel comparators – this is the "fully associative" part of set associativity, but restricted to a small subset of the cache.
4. If any of the N comparators reports a match *and* that entry's **Valid Bit** is 1, it is a **Cache Hit**, and the Offset field selects the requested byte/word from that entry's data.
5. If none of the N comparators matches (or the matching entry's Valid Bit is 0), it is a **Cache Miss**. The requested block is fetched from main memory and placed into the mapped set – into any currently empty way if one is free, or by evicting an existing entry within *that same set only* (chosen according to the cache's replacement policy) if the set is already full.

An N-way set-associative cache therefore needs only N comparators *in total* (reused for whichever set the index selects), rather than one comparator per entry across the whole cache as in a fully associative design – this is precisely why set associativity is described as a compromise between the cheap-but-inflexible direct-mapped scheme and the flexible-but-expensive fully associative scheme.

7.14.3 Worked Example: Full Tag/Index/Offset Calculation

A byte-addressable, **4-way** set associative cache with total volume 2 KB, and a cache line size of 64 bytes, for a 16-bit (64 KB) byte-addressable main memory.

- **Offset**: 64 bytes per line, 1 byte per addressable unit:

$$\log_2(64) = \textbf{6 bits for Offset}$$

- **Set Index**: set size $= 4 \times 64 = 256$ bytes; number of sets $= 2048/256 = 8$:

$$\log_2(8) = \textbf{3 bits for Set Index}$$

- **Tag**: total address length is $\log_2(65536) = 16$ bits, so

$$\text{Tag} = 16 - 3 - 6 = \textbf{7 bits}$$

16 bits (total address)		
Tag Length	Set Index Length	Offset Length
7	3	6

Each of the 8 sets contains 4 ways (cache lines); on any access, the 3-bit Set Index selects exactly one of the 8 sets, and the requested 7-bit Tag is then compared in parallel against only the (up to) 4 tags stored within that selected set.

7.14.4 Advantages and Disadvantages of Set Associative Cache

- **Advantages**: a practical middle ground – substantially reduces the mapping "conflict misses" that plague direct-mapped caches (since N different blocks sharing the same index can now coexist in the cache simultaneously), while needing far fewer comparators than a fully associative cache of the same total size (only N comparators, not one per entry).
- **Disadvantages**: still more complex and expensive than a direct-mapped cache (requires N comparators and some logic to select among matches/empty ways, plus a replacement policy for choosing which way to evict within a full set); still not entirely free of conflict misses, since more than N frequently-used blocks mapping to the same set will still cause thrashing within that set, exactly as a direct-mapped cache would for a single position.
- Increasing N (the associativity) generally reduces conflict misses further, but with diminishing returns and increasing hardware cost – choosing N is itself a design trade-off (common real values are 2-way, 4-way, 8-way, and 16-way set associative caches).

7.15 Cache Replacement Policies

Whenever a set-associative (or fully associative) cache is full and a new block must be brought in, a **replacement policy** decides which existing cache entry to evict. This decision should be made in a way that is well-aligned with the principle of locality being exploited. Common policies include:

- **Least Recently Used (LRU)**: evicts the cache entry that has gone **unused for the longest time**. Directly exploits temporal locality (the assumption that recently used data is likely to be used again soon, so the *least*-recently-used entry is the safest to discard).
- **First In First Out (FIFO)**: evicts the cache entry that has had the **longest period of stay** in the cache, regardless of how recently it was actually used.
- **Least Frequently Used (LFU)**: evicts the cache line with the **lowest access frequency** (fewest total accesses).
- **Random**: evicts a cache entry chosen at random, requiring no extra bookkeeping/state at all.

Advantages and disadvantages (summary):

- **LRU**: closely tracks real usage patterns and generally gives strong performance, but requires extra hardware/bookkeeping (tracking recency for every entry), adding cost and complexity.
- **FIFO**: very simple to implement (just a queue), but can perform poorly since it ignores whether an entry has actually been used recently – it may evict a frequently-used entry simply because it was loaded first.
- **LFU**: can effectively identify genuinely "hot" (popular) data over a long period, but requires maintaining and updating access-frequency counters for every entry, and can be slow to adapt when access patterns change suddenly.
- **Random**: extremely cheap to implement (no bookkeeping at all), and avoids certain pathological worst-case access patterns that can defeat deterministic policies, but does not exploit locality at all, so on average it usually performs worse than LRU.